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Neuroendocrinology and neurobiology of sebaceous glands

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ABSTRACT

The nervous system communicates with peripheral tissues through nerve fibres and the systemic release of hypothalamic and pituitary neurohormones. Communication between the nervous system and the largest human organ, skin, has traditionally received little attention. In particular, the neuro-regulation of sebaceous glands (SGs), a major skin appendage, is rarely considered. Yet, it is clear that the SG is under stringent pituitary control, and forms a fascinating, clinically relevant peripheral target organ in which to study the neuroendocrine and neural regulation of epithelia. Sebum, the major secretory product of the SG, is composed of a complex mixture of lipids resulting from the holocrine secretion of specialised epithelial cells (sebocytes). It is indicative of a role of the neuroendocrine system in SG function that excess circulating levels of growth hormone, thyroxine or prolactin result in increased sebum production (seborrhoea). Conversely, growth hormone deficiency, hypothyroidism, and adrenal insufficiency result in reduced sebum production and dry skin. Furthermore, the androgen sensitivity of SGs appears to be under neuroendocrine control, as hypophysectomy (removal of the pituitary) renders SGs largely insensitive to stimulation by testosterone, which is crucial for maintaining SG homeostasis. However, several neurohormones, such as adrenocorticotropic hormone and α-melanocyte-stimulating hormone, can stimulate sebum production independently of either the testes or the adrenal glands, further underscoring the importance of neuroendocrine control in SG biology. Moreover, sebocytes synthesise several neurohormones and express their receptors, suggestive of the presence of neuro-autocrine mechanisms of sebocyte modulation. Aside from the neuroendocrine system, it is conceivable that secretion of neuropeptides and neurotransmitters from cutaneous nerve endings may also act on sebocytes or their progenitors, given that the skin is richly innervated. However, to date, the neural controls of SG development and function remain poorly investigated and

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incompletely understood. Botulinum toxin-mediated or facial paresis-associated reduction of human sebum secretion suggests that cutaneous nerve-derived substances modulate lipid and inflammatory cytokine synthesis by sebocytes, possibly implicating the nervous system in acne pathogenesis. Additionally, evidence suggests that cutaneous denervation in mice alters the expression of key regulators of SG homeostasis. In this review, we examine the current evidence regarding neuroendocrine and neurobiological regulation of human SG function in physiology and pathology. We further call attention to this line of research as an instructive model for probing and therapeutically manipulating the mechanistic links between the nervous system and mammalian skin.

Key words: skin, sebaceous gland, sebum, nervous system, neurons, neuroendocrinology, hormones, neuropeptides, androgens, acne vulgaris

CONTENTS

I. RELEVANCE OF THE BRAIN–SEBACEOUS GLAND AXIS TO SKIN PHYSIOLOGY AND PATHOLOGY

Disorders of both the peripheral nervous system and neuroendocrine system are associated with cutaneous manifestations, such as the typical oily skin observed with excess growth hormone (GH) secretion, or the muco-cutaneous pigmentation characteristic of adrenal insufficiency in Addison's disease (AD) (Eedy, 2016; Paus, 2016*b*). Long-recognised clinical observations such as these highlight the importance of the nervous system in the regulation of human skin physiology and pathology: a subject of increasing interest in both clinical and experimental dermatology (Slominski & Wortsman, 2000; Paus, Theoharides & Arck, 2006; Misery *et al.*, 2014; Finsterer & Wakil, 2015; Leventhal & Braverman, 2016; Paus, 2016*a*). Yet, in this context, a key appendage of mammalian skin, the sebaceous gland (SG), remains incompletely explored. Careful dissection of the neuroendocrine and neural regulation of SGs promises to improve our understanding of the complex interactions between the nervous system, specialised neuroendocrine glands, and mammalian skin (Arck *et al.*, 2006; Slominski *et al.*, 2012, 2018a, 2018b; Paus, 2016*a*). The current review strives to help close this knowledge gap, and to call attention to the translational research potential awaiting discovery through comprehensive re-examination of SG biology and pathology from a distinct neurobiological and neuroendocrine perspective.

The cells of the SG, known as sebocytes, synthesise and secrete sebum, which is constituted by a diverse mixture of lipids (Picardo *et al.*, 2009; Camera *et al.*, 2010). Evidence from the cutaneous phenotypes present in mice that are deficient for sebaceous lipids (Ehrmann & Schneider, 2016) highlight the important functions of sebum in skin physiology. These functions include: waterproofing of skin and fur (Chen *et al.*, 1997; Chen *et al.*, 2002; Westerberg *et al.*, 2004; Dahlhoff *et al.*, 2016), anti-microbial activity (Basta, Wilburg & Heczko, 1980; Wille & Kydonieus, 2003; Drake *et al.*, 2008; Lee *et al.*, 2009; Nakatsuji *et al.*, 2010), maintenance of skin barrier function (Sundberg *et al.*, 2000; Fluhr *et al.*, 2003; Westerberg *et al.*, 2004; Man *et al.*, 2009), protection against ultraviolet radiation (Beadle & Burton, 1981; Ohsawa *et al.*, 1984; Passi *et al.*, 2002; Mudiyanselage *et al.*, 2003; Ryu *et al.*, 2009; Dahlhoff *et al.*, 2016), and regulation of body temperature (Chen *et al.*, 1997; Chen *et al.*, 2002; Sampath *et al.*, 2009; Dahlhoff *et al.*, 2016).

Furthermore, several skin conditions exhibit SG abnormalities (Shi *et al.*, 2015). Acne vulgaris, or acne, is an exceedingly common skin disease (Tan & Bhate, 2015; Lynn *et al.*, 2016; Rocha & Bagatin, 2018), which is characterised by the formation of comedones (commonly called 'blackheads' or 'whiteheads'). Comedones are cystic dilations of the distal hair follicle (HF) that are associated with atrophic SGs (Kligman, 1974). Generally however, sebum production in those with acne is higher than normal (Cotterill, Cunliffe & Williamson, 1971; Youn *et al.*, 2005; Pappas *et al.*, 2009; Akaza *et al.*, 2014; Bilgiç *et al.*, 2015; Okoro, Bulus & Zouboulis, 2016). Antibiotics can be used to combat inflammatory acne, while retinoids, such as isotretinoin, are effective 'comedolytics', which reduce the number of comedones, lower sebum secretion, and cause SG involution (Dahlhoff, Zouboulis & Schneider, 2016; Kosmadaki & Katsambas, 2017). Atrophic SGs and altered sebaceous lipid composition have also been observed in patients with psoriasis and atopic dermatitis (Prose, 1965; Wirth, Gloor & Stoika, 1981; Burton & Pye, 1983; Werner, Brenner & Böer, 2008; Takahashi *et al.*, 2014; van Smeden *et al.*, 2014; Liakou *et al.*, 2016; Rittié *et al.*, 2016), suggesting that SG dysfunction may contribute to other inflammatory skin diseases (Shi *et al.*, 2015).

Therefore, understanding the regulatory mechanisms that control SG function is clinically important. In recognition of this, the hormonal control of SGs has been under scrutiny since the first half of the 20th century: a focus that stemmed from early observations of increased sebum production and acne prevalence during adolescence (Bloch, 1931). Subsequently, androgens were the first sebotrophic hormones to be discovered, being identified as essential for both the adolescent development of SGs and for maintaining normal sebum production levels in adulthood (Hamilton, 1941; Haskin, Lasher & Rothman, 1953; Pochi & Strauss, 1974). We now know that androgens are also instrumental in acne pathogenesis (Hamilton & Mestler, 1963; Lawrence, Shaw & Katz, 1986; Motosko *et al.*, 2019), and anti-androgens can be used therapeutically to combat acne and excess sebum production (Pye *et al.*, 1977; Miller *et al.*, 1986; Patel & Noble, 1987).

Neurohormones were soon after demonstrated to be required for maintaining sebum production in a series of seminal rodent experiments, where the pituitary glands of rats were surgically removed (hypophysectomy) (1954*b*; Lasher, Lorincz & Rothman, 1954a; Ebling, 1957; Ebling, Ebling & Skinner, 1969a; Ebling *et al.*, 1970; Thody & Shuster, 1971*b*). There is now a wealth of clinical evidence from patients with neuroendocrine disorders associated with cutaneous symptoms that indicate neurohormones directly influence sebum production (Paus, 2016*b*) (Table 1).

Although it has long been known that there is a significant neuroendocrine component to the regulation of SG biology, this topic has yet to be comprehensively reviewed. The fields of neuroendocrinology and neurobiology are conceptually similar, in terms of the release of factors from neurons into the extracellular milieu (Goyal & Chaudhury, 2013). Therefore, we will discuss clinical evidence for neuroendocrine and neurobiological control of SGs alongside experimental data acquired *in vivo*, or through sebocyte culture systems (Ehrmann & Schneider, 2016; Schneider & Zouboulis, 2018; de Bengy *et al.*, 2019). We still make a distinction in our discussions between hormones and regulatory factors typically considered part of the neuroendocrine system, and those associated with the peripheral nervous system [e.g. adrenocorticotropic hormone (ACTH) versus acetylcholine].

II. INTRODUCTION TO SEBACEOUS GLAND BIOLOGY

(1) Microanatomy of the pilosebaceous unit

The SG is connected to the distal portion of the HF *via* a keratinised duct, through which sebum is released (Figs 1 and 2A). Secreted sebum coats the hair shaft, pilary canal and the surface of the skin. Pilosebaceous units (PSUs) consist of HFs and SG lobes arranged in close proximity, along with an arrector pili muscle and, in regions of skin that contain them, an apocrine gland (AG) (Hinde *et al.*, 2013; Schneider & Paus, 2014) (Fig. 1 and 2A). Furthermore, in humans, the coil of an eccrine gland can also be found adjacent to the HF above the bulbar region, often together with a cone of dermal adipocytes that enwrap the bulge in human scalp HFs (Poblet *et al.*, 2018). The region where the SG duct inserts into the pilary canal is referred to as the junctional zone (JZ). Distal to the JZ is the HF infundibulum, where the hair shaft separates from the follicular epithelium (Fig. 1). The entire PSU is encapsulated by a connective

Table 1. Conditions of the central neuroendocrine system linked to altered sebum production

Condition	Implicated hormone(s)	Sebum production	Reference(s)
Addison's disease	↓CRH, ACTH, or corticosteroids	↓sebum	Pochi, Strauss & Mescon (1963)
Acromegaly	↑GH	↑sebum	Burton *et al.* (1972); Borlu *et al.* (2012)
Parkinson's disease	↑PRL, ↓dopamine	↑sebum	Pochi, Strauss & Mescon (1962a); Burton *et al.* (1973,*b*); Binder & Jonelis (1983); Cotterill *et al.* (1971); Villares & Carlini (1989); Martignoni *et al.* (1997)
Phenylketonuria	↑PRL, ↓dopamine	↑sebum	Burton, Goolamali & Shuster (1975); Schulpis *et al.* (1998); Denecke *et al.* (2000); van Spronsen (2010); Juhász *et al.* (2016)
Pregnancy, breastfeeding	↑PRL	↑sebum	Burton *et al.* (1970); Shuster & Thody (1974); Burton, Shuster & Cartlidge (1975); Henderson, Taylor & Cunliffe (2000)
Hypothyroidism	↓TRH, TSH, or T_3/T_4	↓sebum	Goolamali, Evered & Shuster (1976)
Growth hormone deficiency	↓GH	↓sebum	Goolamali, Burton & Shuster (1973); Pochi & Strauss (1974); Tanriverdi *et al.* (2006); Borlu *et al.* (2007)

ACTH, adrenocorticotropic hormone; CRH, corticotropin-releasing hormone; GH, growth hormone; PRL, prolactin; TRH, thyrotropin-releasing hormone; TSH, thyroid-stimulating hormone; T3/T4, thyroid hormones.
Arrows indicate increased/decreased levels of hormones or sebum production for a given condition.

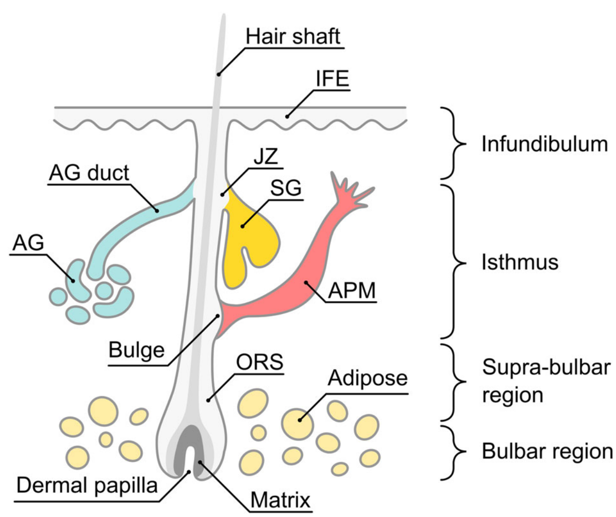

Fig. 1. Anatomy of the human pilosebaceous unit (PSU), comprised of the hair follicle, sebaceous gland (SG) and, in regions of skin that contain them, the apocrine gland (AG). APM, arrector pili muscle; IFE, interfollicular epidermis; JZ, junctional zone; ORS, outer root sheath.

tissue sheath (CTS), or stroma, made up of mesenchymal fibroblasts and collagen (Hinde *et al.*, 2013; Schneider & Paus, 2014).

(2) Development of the sebaceous gland

Human SGs appear around 3.5 months into the gestational period (Kligman & Strauss, 1956; Montagna & Parakkal, 1974*b*). However, in mice, the first sebocytes emerge in the late-foetal/early-postnatal period, during the 'bulbous peg stage' of PSU development (Paus *et al.*, 1999; Frances & Niemann, 2012). Prior to the appearance of sebocytes, a discrete population of cells that express both leucine-rich repeats and immunoglobulin-like domains protein 1 (LRIG1) and SRY-box transcription factor 9 (SOX9) forms within the developing hair peg. As development of the PSU proceeds, two distinct groups of cells emerge that differentially express either LRIG1 or SOX9, and become restricted to the JZ and hair bulge compartments, respectively. The first sebocytes emerge from the LRIG1-expressing population in the JZ (Nowak *et al.*, 2008; Frances & Niemann, 2012; Schepeler, Page & Jensen, 2014; Xu *et al.*, 2015; Ouspenskaia *et al.*, 2016). Human SGs are active *in utero*, as at least part of the vernix caseosa of neonates is derived from sebaceous secretions (Rissmann *et al.*, 2006; Míková *et al.*, 2014). SGs are then relatively inactive throughout childhood, increasing in size and activity only once adolescence is reached (Pochi, Strauss & Mescon, 1962b; Cotterill *et al.*, 1972; Montagna & Parakkal, 1974*b*).

(3) Structure and cellular homeostasis of the sebaceous gland

The SG has a layered structure that corresponds to different stages of sebocyte differentiation (Figs 2 and 3). The outermost sebocytes make up the basal layer or peripheral zone (PZ), and are relatively undifferentiated and proliferative (Epstein & Epstein, 1966; Hamilton, 1974; Hamilton, Howard & Potten, 1974) (Figs 2A, D and 3). Much of what is known about the factors that drive sebocyte mitosis and differentiation has been ascertained using transgenic mice (Ehrmann & Schneider, 2016; Clayton *et al.*, 2019). Sebocyte proliferation is promoted by multiple factors such as C-myc (Horsley *et al.*, 2006; Jensen *et al.*, 2009; Cottle *et al.*, 2013), androgens (Ebling, Ebling & Skinner, 1969a; Petersen,

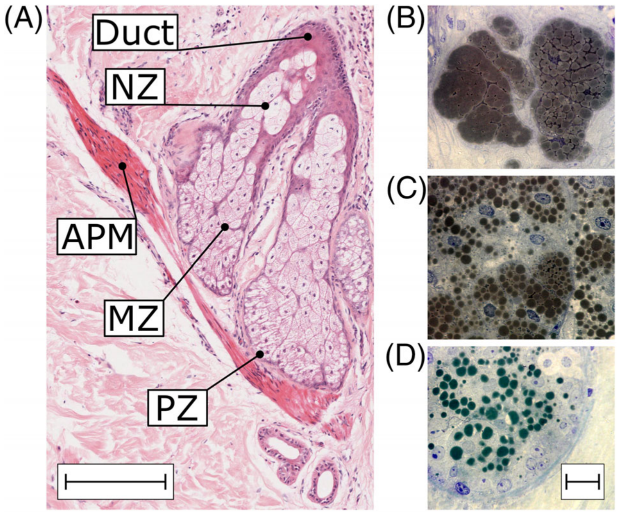

Fig. 2. Histology of the human sebaceous gland (SG). (A) Histomicrograph of a haematoxylin and eosin-stained SG, arrector pili muscle (APM), and surrounding stroma and dermis. Scale is equivalent to 200 μm. (B–D) Histomicrographs of OsO_4- and toluidine blue-stained SGs showing different stages of sebocyte differentiation. Scale in each panel is equivalent to 20 μm. (B) Terminally differentiated sebocytes comprise the necrotic zone (NZ) where they undergo holocrine secretion, releasing sebum into the SG duct. (C) The maturation zone (MZ) contains differentiating sebocytes, evident by lipid droplet accumulation. (D) The peripheral zone (PZ) contains undifferentiated sebocytes.

Zone & Krueger, 1984; Akamatsu, Zouboulis & Orfanos, 1992; Fujie *et al*., 1996; Rosignoli *et al*., 2003), and receptor tyrosine kinases including epidermal growth factor receptor (EGFR) and related receptors (Bol *et al*., 1998; Kiguchi *et al*., 2000; Sato *et al*., 2001; Akimoto *et al*., 2002; Dahlhoff *et al*., 2013*b*; Li *et al*., 2016), as well as fibroblast growth factor receptor 2b (Fgfr2b) (Danilenko *et al*., 1995). Recently, caspase 3-mediated activation of yes-associated protein (YAP) was demonstrated to promote mitosis in murine SGs (Yosefzon *et al*., 2018).

Adjacent to the peripheral layer is the supra-basal layer, or maturation zone (MZ), which contains differentiating sebocytes (Figs 2A, C and 3). Differentiation is evident by the accumulation of lipid droplets in sebocytes within the MZ, and dramatically increased cell size compared to the PZ (Figs 2A, C, D and 3). Finally, the necrotic zone (NZ) is where terminally differentiated sebocytes release sebum by undergoing holocrine secretion, a process that resembles autophagic cell death (Brandes, Bertini & Smith, 1965; Mesquita-Guimarães & Coimbra, 1976; Fischer *et al*., 2017; Zouboulis, 2017; Rossiter *et al*., 2018) (Figs 2A, B and 3). The time taken for an undifferentiated sebocyte in the peripheral layer to differentiate fully is around 7–14 days, and is comparable between human and rodent SGs (Epstein & Epstein, 1966; Hamilton, 1974; Hamilton, Howard & Potten, 1974; Jung *et al*., 2015).

Lipid synthesis is central to differentiation in the SG. This is reflected in the importance of transcription factors that promote the expression of lipogenic enzymes to sebocyte differentiation. Such transcription factors include peroxisome proliferator-activated receptor gamma (PPARγ) (Berger & Moller, 2002; Icre, Wahli & Michalik, 2006) and CCAAT-enhancer-binding proteins (C/EBPs) (Chung *et al*., 2012; Guo, Li & Tang, 2015), both of which are required for sebum production and SG maintenance (Karnik *et al*., 2009; House *et al*., 2010; Sardella *et al*., 2017). Androgens, through adrenoceptors (ARs), also promote sebocyte differentiation (Ebling, Ebling & Skinner, 1969a; Petersen, Zone & Krueger, 1984; Rosignoli *et al*., 2003; Cottle *et al*., 2013). Broad manipulation of wingless (Wnt) signalling throughout the murine PSU reveals that low levels of Wnt promote sebaceous differentiation (Gat *et al*., 1998; Merrill *et al*., 2001; Niemann *et al*., 2002, 2007; Collins & Watt, 2008; Petersson *et al*., 2011; Kakanj *et al*., 2013; Lien *et al*., 2014), while high levels of Wnt signalling result in SG loss and ectopic HF formation (Lo Celso, Prowse & Watt, 2004; Kretzschmar *et al*., 2016). Conversely, hedgehog signalling may act to promote sebocyte differentiation (Allen *et al*., 2003; Oro & Higgins, 2003; Gu & Coulombe, 2008).

(4) Sebaceous gland progenitor cells

Terminal differentiation of sebocytes results in holocrine secretion and cell death. Therefore, a continuous supply of new sebocytes is required. Lineage tracing studies in mice have demonstrated that a population of Lrig1-expressing cells in the JZ of the PSU generates both the SG and infundibulum, and likely constitutes a progenitor cell population for both compartments (Nowak *et al*., 2008; Jensen *et al*., 2009; Barnes, Saurat & Kaya, 2017) (Fig. 3B). LRIG1 also labels undifferentiated cells in human Meibomian glands (Xie *et al*., 2018), which are a specialised form of SG present in the eyelid (Butovich, 2017). Ectopic activation of Wnt signalling in Lrig1+ cells results in dichotomous expansion of the upper HF and loss of SGs (Kretzschmar *et al*., 2016). This observation, combined with the described formation of sebaceous tumours in Wnt signalling-deficient mouse skin (Merrill *et al*., 2001; Niemann *et al*., 2002, 2007; Petersson *et al*., 2011; Lien *et al*., 2014), would suggest that Wnt signalling controls the lineage commitment decisions of progenitors for the SG and infundibulum.

Within the murine SG, cells that express leucine-rich repeat containing G protein-coupled receptor 6 (Lgr6) generate the SG predominantly through stochastic cell division (Snippert *et al*., 2010; Kretzschmar *et al*., 2014; Füllgrabe *et al*., 2015). Other Lgr6+ populations can be found in the HF and interfollicular epidermis (IFE), each contributing to its respective compartment largely independently (Kretzschmar *et al*., 2014; Füllgrabe *et al*., 2015). Similar to the phenotype of ectopic Wnt signalling in the Lrig1+ compartment, expression of constitutively active β-catenin in Lgr6-expressing cells leads to loss of SGs and expansion of the upper HF (Kretzschmar *et al*., 2016).

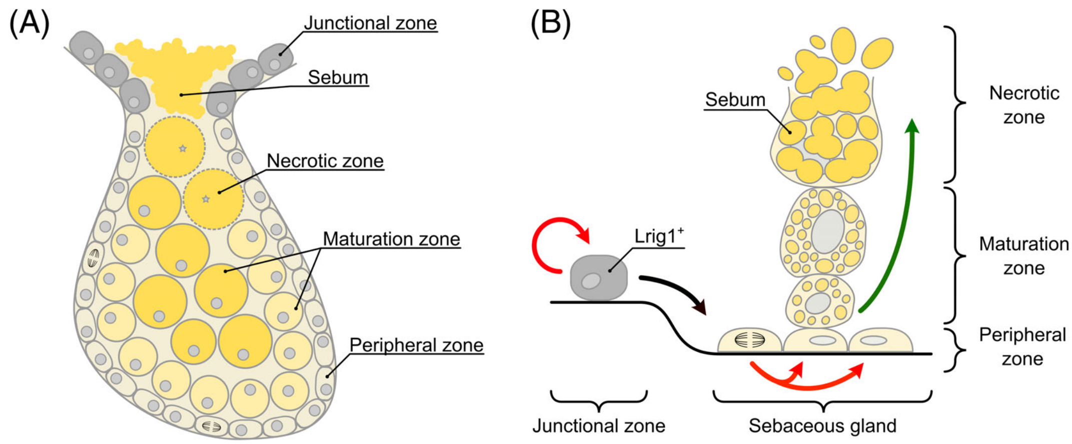

Fig. 3. Structure and cellular homeostasis of sebaceous glands (SGs). (A) Diagrammatic representation of overall SG structure and compartmentalisation into zones corresponding to different stages of sebocyte differentiation. (B) Schematic of cellular homeostasis of SGs. Stem cells in the junctional zone that express leucine-rich repeats and immunoglobulin-like domains protein 1 (Lrig1) give rise to the SG. Proliferation occurs chiefly within the peripheral zone. Undifferentiated sebocytes in the peripheral layer differentiate by synthesising lipids and moving into the maturation zone. Terminal differentiation and cell death by holocrine secretion occurs in the necrotic zone. Red arrows indicate cell division. The black arrow indicates translocation of cells. The green arrow indicates differentiation.

(5) Sebum composition

Sebum is composed of several different types of lipids, and sebum composition varies considerably among different species (Thody & Shuster, 1989; Smith & Thiboutot, 2008). In human sebum, triglycerides (TGs), squalene, and wax esters predominate, although cholesterol and cholesterol esters also constitute a small fraction (Summerly & Woodbury, 1971, 1972; Summerly *et al.*, 1976; Ridden, Ferguson & Kealey, 1990; Guy, Ridden & Kealey, 1996; Picardo *et al.*, 2009). While sebum on the surface of the skin contains a significant proportion of free fatty acids (FFAs) (Downing, Strauss & Pochi, 1969; Greene *et al.*, 1970; Nicolaides *et al.*, 1970), analysis of lipid composition from isolated SGs reveals that FFAs comprise only a comparatively small fraction of lipid synthesised in the SG (Summerly & Woodbury, 1971, 1972; Ridden, Ferguson & Kealey, 1990; Guy, Ridden & Kealey, 1996). This discrepancy may be caused by lysis of TGs by commensal microorganisms including *Cutibacterium acnes* (*C. acnes*, formerly known as *Propionibacterium acnes*) and yeasts of the genus *Malassezia* (Marples, Downing & Kligman, 1971; Puhvel, Reisner & Sakamoto, 1975; Scholz & Kilian, 2016; Sommer *et al.*, 2016).

Human sebum is further characterised by the presence of unique lipids including squalene and $\Delta 6$ monounsaturated fatty acids like sapienic acid (Nicolaides, 1974; Thody & Shuster, 1989). Lipid droplet-associated proteins, such as perilipins, regulate the size and rate of formation of lipid droplets (Dahlhoff *et al.*, 2013*a*, 2015; Camera *et al.*, 2014; Schneider, 2015*b*; Schneider, Zhang & Li, 2016). Interestingly, sebum composition varies by anatomical region, corresponding to differences in SG number (Ludovici *et al.*, 2018) and SG size (Summerly *et al.*, 1976). Moreover, sebum composition is often altered in diseased skin (Shi *et al.*, 2015). In acne, for example, there is an increased presence of squalene, monounsaturated fatty acids, and diaglycerols (Cotterill *et al.*, 1972; Pappas *et al.*, 2009; Akaza *et al.*, 2014; Camera *et al.*, 2016), while psoriasis and atopic dermatitis both feature a reduction in total FFAs and FFA chain length (Takahashi *et al.*, 2014; van Smeden *et al.*, 2014). Altered sebum composition might also underlie skin hypersensitivity to allergens (Fan *et al.*, 2018). However, the direct functional importance of sebum composition remains untested and is incompletely understood.

III. NEUROENDOCRINE REGULATION OF SEBACEOUS GLAND FUNCTION: OVERVIEW

There is substantial clinical evidence indicating that neurohormones are involved in regulating SG function (Table 1). For example, hyperprolactinemia is associated with markedly increased levels of sebum production (Pochi, Strauss & Mescon, 1962a; Cotterill *et al.*, 1971; Burton *et al.*, 1973; Binder & Jonelis, 1983; Juhász *et al.*, 2016). Patients with acromegaly or growth hormone deficiency exhibit increased and decreased sebum production, respectively (Burton *et al.*, 1972; Goolamali, Burton & Shuster, 1973; Pochi & Strauss, 1974; Tanriverdi *et al.*, 2006; Borlu *et al.*, 2007, 2012). Furthermore, psycho-emotional stress activates the hypothalamic–pituitary–adrenal (HPA) axis (Miller & O'Callaghan, 2002), and exacerbation of acne and seborrhoeic dermatitis by stress has long been appreciated (Chiu, Chon & Kimball, 2003; Misery *et al.*, 2007; Yosipovitch *et al.*, 2007; Orion & Wolf, 2014; Dréno, 2017). Conversely, hypo-function of the HPA axis, such as in Addison's disease, results in reduced sebum production (Pochi, Strauss & Mescon, 1963). The challenge has been, and remains, to systematically follow up these clinical pointers experimentally, and to identify the specific neurohormones involved. Research to this end has made extensive use of *in vitro* sebocyte and SG culture systems, and *in vivo* data from rodent models.

The seminal experimental observation that hypophysectomy resulted in reduced sebum secretion in rodents first pointed to the neuroendocrine regulation of the skin (Lasher, Lorincz & Rothman, 1954a, 1954b; Ebling, 1957; Ebling, Ebling & Skinner, 1969b; Ebling *et al.*, 1970; Thody & Shuster, 1971*b*) (Fig. 4). Crucially, where exogenous testosterone restores sebum secretion in castrated animals, such recovery is profoundly impaired in animals that have been both castrated and hypophysectomised (Lasher, Lorincz & Rothman, 1954a, 1954b; Ebling, 1957; Ebling, Ebling & Skinner, 1969b; Ebling *et al.*, 1970; Thody & Shuster, 1971*b*) (Fig. 4). These observations suggest that the well-known stimulatory action of testosterone on the SGs depends on the presence of other pituitary-derived hormones. One hypothesis as to the mechanism behind this phenomenon is that neurohormones act on sebocytes to modulate the enzymatic processing of testosterone to more active metabolites, such as dihydroandrosterone (DHT) (Randall & Ebling, 1982; Ebling, 1986). Indeed, while testosterone has little effect in castrated–hypophysectomised rats, DHT remains an effective stimulator of sebum secretion, despite the absence of the pituitary (Ebling *et al.*, 1971, 1973).

Certain neurohormones, such as α-melanocyte-stimulating hormone (α-MSH) and possibly GH, exert their own testosterone-independent sebotrophic effect, whilst simultaneously facilitating the stimulatory action of testosterone on rat SGs (Ebling, Ebling & Skinner, 1969b; Ebling *et al.*, 1975a, 1975b). However, ACTH stimulates sebum production while failing to influence the sebaceous response to testosterone (Ebling *et al.*, 1970; Thody & Shuster, 1971*a*) (Fig. 4). In the more than 40 years that have passed since the last of these seminal experiments, very little further research has been conducted on the regulation of SGs by neurohormones *in vivo*.

IV. THE HYPOTHALAMIC–PITUITARY–ADRENAL AXIS AND MELANOCORTIN SYSTEM

(1) Overview of the hypothalamic-pituitary-adrenal axis and melanocortin system

The external factors that result in activation of the hypothalamic–pituitary–adrenal (HPA) axis can be collectively termed ‘stress’. These include psychological stressors and other factors such as infection or injury (Slominski, Ermak & Mihm, 1996; Slominski & Mihm, 1996; Arck *et al.*, 2006; Slominski *et al.*, 2007). The HPA axis is comprised of several neurohormones and steroid hormones. Corticotropin-releasing hormone (CRH) is released by neurons of the hypothalamic paraventricular nucleus into the hypophyseal blood supply, where it stimulates the release of ACTH from the corticotropic cells of the anterior pituitary. ACTH subsequently activates cells within the adrenal cortex, causing the systemic release of adrenal steroid hormones (Aguilera, 1998; Smith & Vale, 2006) (Fig. 5).

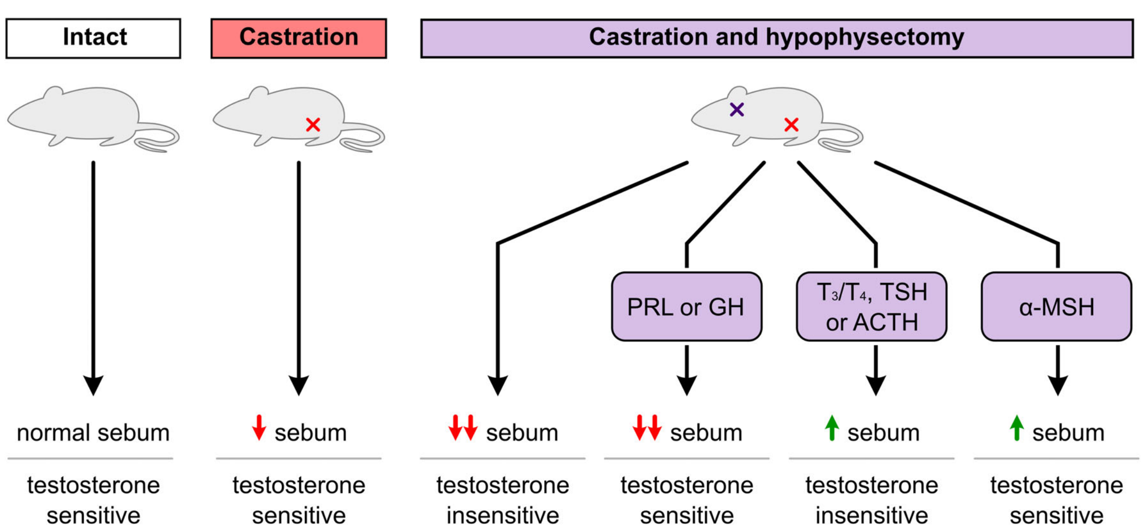

Fig. 4. Effects of castration and hypophysectomy (removal of pituitary gland) on sebum production in rats. Castration results in reduced sebum production that can be rescued by administering exogenous testosterone. However, concomitant castration and hypophysectomy results in an impaired ability to rescue sebum production with testosterone. Sensitivity to testosterone can be restored in castrated–hypophysectomised rats using several pituitary neurohormones including prolactin (PRL) and growth hormone (GH) which have a limited effect on sebum production on their own. Thyroid hormones T_3 and T_4, thyroid-stimulating hormone (TSH) and adrenocorticotropic hormone (ACTH), when administered, increase sebum production but do not restore sensitivity to testosterone. Both sebum production and sensitivity to testosterone are positively influenced by administration of α-melanocyte-stimulating hormone (α-MSH). Red arrows indicate decreased sebum levels relative to intact rats. Green arrows indicate increased levels of sebum relative to castrated–hypophysectomised rats.

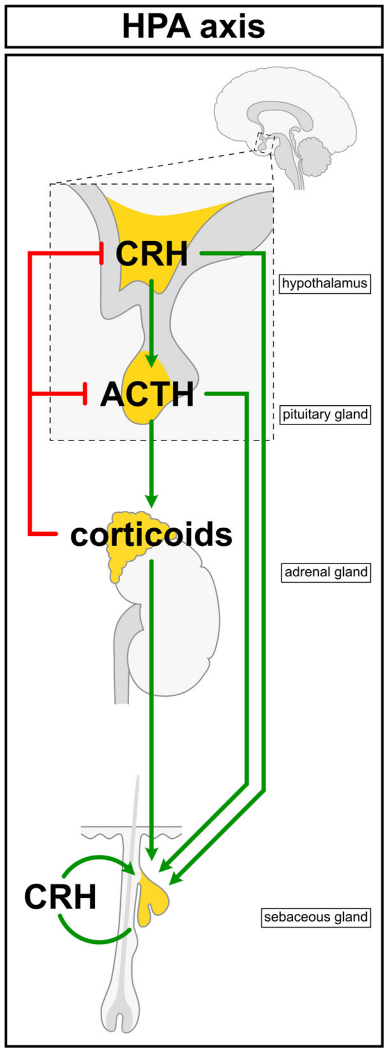

Fig. 5. Control of sebaceous glands (SGs) by the hypothalamic–pituitary–adrenal (HPA) axis. Corticotropin-releasing hormone (CRH), produced by the hypothalamus, stimulates release of adrenocorticotropic hormone (ACTH) from corticotropic cells in the anterior pituitary. In turn, ACTH acts on the adrenal gland, causing release of corticoids, such as cortisol, which participate in a negative feedback loop to reduce CRH and ACTH secretion. Hypothalamic CRH and pituitary-derived ACTH may act directly on SGs, in addition to indirect stimulation by mediating production of adrenal hormones. SG-derived CRH might also act as an autocrine hormone to regulate SG function (see Fig. 6).

The receptor proteins for CRH and ACTH are all G protein-coupled receptors (GPCRs). CRH has two receptors, CRHR1 and CRHR2 (Grammatopoulos, 2012; Slominski *et al.*, 2013), while the primary receptor for ACTH is the MC2R melanocortin receptor (Yang, 2011). However, ACTH can also bind and activate other melanocortin receptors, of which there are five in total (MC1–5R) (Yang, 2011). The HPA axis shares these receptors with the melanocortin system (MCS), which regulates pigmentation and energy homeostasis (Ellacott & Cone, 2006). In addition to ACTH, which also acts as a melanocortin (Hunt *et al.*, 1994), the MCS deploys several other neurohormones, including the melanocyte-stimulating hormones, α-MSH, β-MSH, and γ-MSH. ACTH and the MSHs are derived *via* proteolytic cleavage of the proopiomelanocortin (POMC) pre-peptide, which is predominantly synthesised in the arcuate nucleus of the hypothalamus and pituitary (Gantz & Fong, 2003).

Aside from the central HPA axis, it is now appreciated that the skin expresses several HPA axis components (Slominski *et al.*, 2001, 2004, 2013). For example, CRH and CRHRs are synthesised in both human skin and scalp HFs (Slominski *et al.*, 1996*a*, 1998, 2000*d*; Quevedo *et al.*, 2001; Zbytek *et al.*, 2002; Ito *et al.*, 2005; Zbytek & Slominski, 2005; Ganceviciene *et al.*, 2009), and in mouse skin (Slominski *et al.*, 1996*b*; Roloff *et al.*, 1998). Human skin also contains urocortin, which is a CRH-related protein and ligand for CRH receptors (Slominski *et al.*, 2000*b*). Skin CRH receptors undergo alternative splicing (Pisarchik & Slominski, 2001, 2004; Slominski *et al.*, 2006*b*) and couple alternate signalling transduction mechanisms in different cell types (Fazal *et al.*, 1998; Zbytek, Pfeffer & Slominski, 2003; Zbytek, Pfeffer & Slominski, 2004; Slominski *et al.*, 2006a, 2006b).

Importantly, both human and mouse SGs have been shown to express several key ligands and receptors of both the HPA axis and melanocortin system (see Section IV.3; Table 2) (Kono *et al.*, 2001; Zhang *et al.*, 2006; Ganceviciene *et al.*, 2009; Eisinger *et al.*, 2011), suggesting that not only is the SG competent to respond to the output of these central neuroendocrine axes, but that peripheral equivalents in skin may act as autocrine regulators of SG function.

(2) Reduced sebum secretion in Addison's disease and following adrenalectomy

Sebum production in humans correlates with systemic levels of adrenal cortisone and hydrocortisone (Strauss & Kligman, 1959; Undeutsch, 1962; Pochi, Strauss & Mescon, 1962b, 1963; Goolamali *et al.*, 1974). When administered to healthy individuals, both ACTH and hydrocortisone produce an enlargement of SGs and outbreaks of acne lesions (Strauss & Kligman, 1959). Indeed, use of anabolic steroids is well recognised as a clinically relevant cause of acne (Horwitz, Andersen & Dalhoff, 2019). Conversely, adrenalectomy (surgical removal of the adrenal glands) reduces sebum secretion levels, and can also reduce acne severity (Pochi, Strauss & Mescon, 1963; Stratakis *et al.*, 1998).

The effects of bilateral adrenalectomy can be attributed, at least in part, to the loss of adrenal steroid hormones (Strauss & Kligman, 1959; Undeutsch, 1962; Pochi, Strauss & Mescon, 1962b, 1963; Goolamali *et al.*, 1974). However, patients suffering from adrenal insufficiency that is secondary to an underlying hypopituitarism demonstrate an impaired recovery to normal levels of sebum secretion following hydrocortisone replacement therapy. This is in contrast to the comparatively good recovery of sebum secretion seen in patients that exhibit adrenal insufficiency only (Goolamali, Burton &

Table 2. Expression of neuroendocrine system components in sebaceous glands (SGs)

Component	Associated system or axis	Description	Species/system	Methodology	Reference(s)
CRH	HPA	Hypothalamic neurohormone	Human SGs	ISH, IHC	Kono *et al.* (2001); Kim *et al.* (2006); Ganceviciene *et al.* (2009)
			SZ95 sebocytes	PCR, WB	Zouboulis *et al.* (2002)
CRH-BP	HPA	CRH-binding protein	Human SGs	IHC	Ganceviciene *et al.* (2009); Elewa *et al.* (2012)
			SZ95 sebocytes	PCR, WB	Zouboulis *et al.* (2002)
CRHR-1	HPA	GPCR, CRH receptor	Human SGs	IHC	Ganceviciene *et al.* (2009); Elewa *et al.* (2012)
			SZ95 sebocytes	PCR, WB	Zouboulis *et al.* (2002)
CRHR-2	Receptor, HPA	GPCR, CRH receptor	Human SGs	IHC	Ganceviciene *et al.* (2009); Elewa *et al.* (2012)
			SZ95 sebocytes	PCR, WB	Zouboulis *et al.* (2002)
POMC	Melanocortin, HPA	Prohormone	Human SGs	LC-PCR, ISH	Kono *et al.* (2001)
			Mouse SGs	ISH, IHC	Mazurkiewicz, Corliss & Slominski (2000)
β-endorphin	Opioid	Opioid, POMC-derived	Mouse SGs	ISH, IHC	Slominski, Paus & Mazurkiewicz (1992); Furkert *et al.* (1997); Mazurkiewicz, Corliss & Slominski (2000)
MC1R	Melanocortin	GPCR, melanocortin receptor	Human SGs	IF	Böhm *et al.* (2002); Zhang *et al.* (2006); Whang *et al.* (2011)
			Primary human sebocytes	PCR	Zhang *et al.* (2003, 2006); Eisinger *et al.*, 2011
			SZ95 sebocytes	PCR	Böhm *et al.* (2002)
MC2R	Melanocortin	GPCR, melanocortin receptor	Human SGs	IF, IHC	Hong-Wei *et al.* (2010)
MC5R	Melanocortin	GPCR, melanocortin receptor	Human SGs	IF	Zhang *et al.* (2006)
			Primary human sebocytes	PCR	Zhang *et al.* (2003, 2006); Eisinger *et al.* (2011)
			Mouse SGs	ISH	Chen *et al.* (1997)

CRH, corticotropin-releasing hormone; GPCR, G-protein-coupled receptor; HPA, hypothalamic-pituitary-adrenal axis; IF, immunofluorescence; IHC, immunohistochemistry; ISH, in situ hybridisation; LC-PCR, laser-capture PCR; PCR, polymerase chain reaction; POMC, proopiomelanocortin; WB, western blot.

Shuster, 1973). Therefore, HPA-axis neurohormones may be required for priming SGs to respond to adrenal androgens and corticosteroids. Alternatively, they may be able to contribute to sebum production independently of the adrenal glands.

(3) Melanocortins and corticotropin-releasing hormone act directly on sebocytes

Administration of ACTH to rats results in an increase in both SG size and sebum excretion (Haskin, Lasher & Rothman, 1953; Ebling *et al.*, 1970; Thody & Shuster, 1970*a*; Ozegović & Milković, 1972). Importantly, these effects still manifest in adrenalectomised rats, indicating that ACTH is able to stimulate SGs independently of adrenal secretions, and therefore independently of the central HPA axis (Thody & Shuster, 1970*a*; Ozegović & Milković, 1972). Furthermore, the action of ACTH does not depend on the presence of the testes, nor does ACTH work synergistically with concomitantly administered testosterone (Ebling *et al.*, 1970) (Fig. 4). Like ACTH, α-MSH also stimulates sebum secretion in rats, and can do so independently of the pituitary gland and testes (Thody & Shuster, 1975; Ebling *et al.*, 1975*b*; Thody *et al.*, 1976; Thody, Meddis & Shuster, 1978) (Fig. 4). However, unlike ACTH, α-MSH exhibits a synergistic effect on sebum secretion when simultaneously administered with testosterone (Fig. 4) (Thody & Shuster, 1975; Ebling *et al.*, 1975*b*; Thody *et al.*, 1976).

In support of the model of direct action of ACTH and α-MSH on sebocytes, the melanocortin receptors MC1R (Böhm *et al.*, 2002; Zhang *et al.*, 2003, 2006; Eisinger *et al.*, 2011; Whang *et al.*, 2011), MC2R (Hong-Wei *et al.*, 2010) and MC5R (Zhang *et al.*, 2003, 2006, 2011; Eisinger *et al.*, 2011) are expressed in human SGs (Table 2). MC5R is also present in murine SGs (Chen *et al.*, 1997; Thiboutot *et al.*, 2000). Mice that are genetically deficient for Mc5r demonstrate impaired temperature homeostasis and a reduced ability to shed water from their fur, similar to other sebaceous lipid-deficient mice (Chen *et al.*, 1997; Sampath *et al.*, 2009; Dahlhoff *et al.*, 2016). Importantly, Mc5r$^{-/-}$ mice exhibit a 15–20% reduction in sebum production (Chen *et al.*, 1997).

ACTH and α-MSH both induce lipid droplet formation in primary human sebocytes (Zhang *et al.*, 2003). In human skin xenografts and primary human sebocytes, application of specific MC1R and MC5R inhibitors results in reduced lipid production and reduced SG size (Eisinger *et al.*, 2011).

Furthermore, human SGs and SZ95 sebocytes express the receptors for CRH (CRH-R1 and CRH-R2), as well as CRH and CRH-binding protein (CRH-BP) as indicated in Table 2 (Kono *et al.*, 2001; Zouboulis *et al.*, 2002; Kim *et al.*, 2006; Ganceviciene *et al.*, 2009; Elewa *et al.*, 2012). Peri-follicular nerve fibres reportedly also synthesise CRH (Roloff *et al.*, 1998; Slominski *et al.*, 1999). Intriguingly, immunoreactivity for CRH is increased throughout the SG in acne lesions (Ganceviciene *et al.*, 2009) and CRH expression can be induced in human keratinocytes through incubation with bacterial lipopolysaccharide (Zbytek & Slominski, 2007), suggesting that the changes in the skin microbiome observed in acne (Fitz-Gibbon *et al.*, 2013; Xu & Li, 2019) might impact on sebocyte CRH expression.

When applied to SZ95 sebocytes, CRH induces synthesis of both neutral and polar lipids, inhibits proliferation, and stimulates the release of pro-inflammatory interleukins IL-6 and IL-8 (Zouboulis *et al.*, 2002; Krause *et al.*, 2007). These findings are in line with the model of CRH acting as a pro-inflammatory mediator in skin (Slominski, 2009; Slominski *et al.*, 2013). Moreover, the anti-proliferative effects of CRH on sebocytes mirror the known inhibitory effect of CRH on human scalp HF growth *ex vivo* (Ito *et al.*, 2005), and can be ameliorated in SZ95 sebocytes using antagonists specific for CRH receptors such as anatalarmin and α-helical CRH (Krause *et al.*, 2007). At a concentration of 100 pM, CRH also produces a significant increase in the mRNA for 3β-hydroxysteroid dehydrogenase, which is a key enzyme involved in androgen and progesterone synthesis (Zouboulis *et al.*, 2002).

The expression of both CRH receptors and CRH ligand within the SG would suggest that CRH can act as an autocrine hormone, regulating lipid synthesis and inflammation (Table 2; Fig. 6). Similarly, POMC has been reported to be synthesised within peripheral sebocytes of human SGs (Table 2) (Kono *et al.*, 2001) and mouse SGs exhibit immuno-reactivity for the POMC-derived endogenous opioid, β-endorphin (Slominski, Paus & Mazurkiewicz, 1992; Furkert *et al.*, 1997). Moreover, as mentioned previously, mammalian skin, including human epidermis and scalp HFs, contain peripheral equivalents of the central HPA axis (Slominski & Wortsman, 2000; Slominski *et al.*, 2000*c*, 2007; Ito *et al.*, 2005; Paus *et al.*, 2014) (Fig. 6). However,

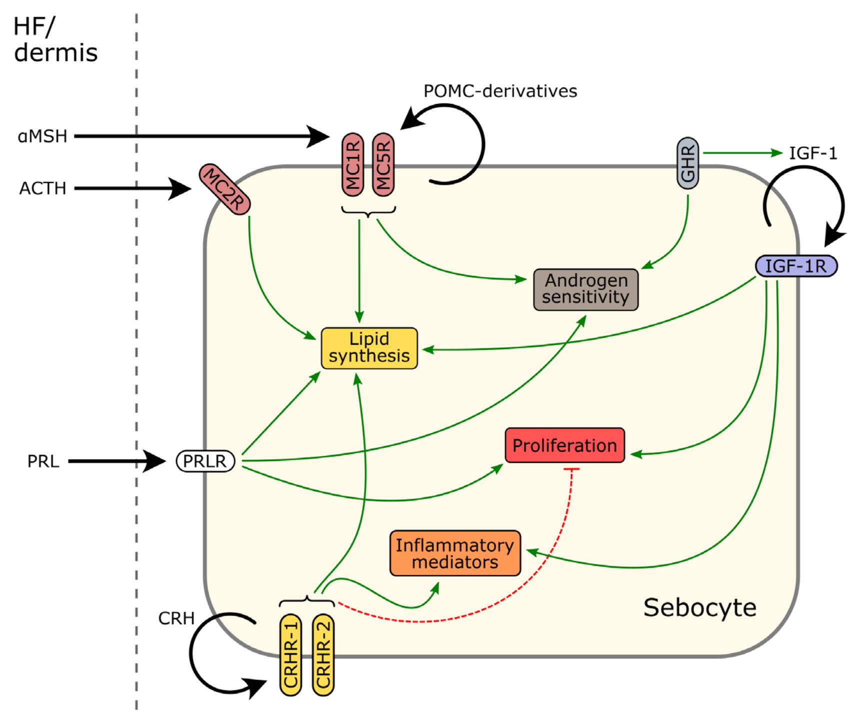

Fig. 6. Regulation of sebocyte biology by intra-cutaneous neuroendocrine system equivalents. Sebocytes express several neurohormones and their cognate receptors, indicating that these neurohormones may act *via* autocrine mechanisms to modulate sebocyte function. It is also known that the hair follicle (HF) and the dermis express multiple neurohormones which could act on sebaceous glands (SGs) in a paracrine fashion. Secretion and ligand binding are indicated by thick black arrows. Implicated stimulatory effects of receptors are indicated by green arrows, while inhibitory effects are shown using dashed red lines. α-MSH, α-melanocyte-stimulating hormone; ACTH, adrenocorticotropic hormone; CRH, corticotropin-releasing hormone; CRHR-1, CRH receptor 1; CRHR-2, CRH receptor 2; GHR, growth hormone receptor; IGF-1, insulin-like growth factor-1; IGF-1R, IGF-1 receptor; MC1R, melanocortin 1 receptor; MC2R, melanocortin 2 receptor; MC5R, melanocortin 5 receptor; POMC, pro-opiomelanocortin; PRL, prolactin; PRLR, prolactin receptor.

the functional importance of these skin-derived neurohormones to SG biology, *versus* those produced by the central nervous system, is not yet known.

V. PROLACTIN AND DOPAMINE

(1) Prolactin and sebum during pregnancy and breastfeeding

Prolactin (PRL) is a pituitary neurohormone, best recognised for its role in promoting lactation in mammals (Horseman & Gregerson, 2014). Typically, circulating PRL levels are higher in women, although this may depend on the menstrual cycle (Tyson *et al*., 1972; Yamaji *et al*., 1976; Farland *et al*., 2017; Langan, 2018). During pregnancy, levels of PRL are increased and peak near term. Circulating levels of PRL decrease postpartum, but further PRL release from the pituitary can be stimulated *via* breastfeeding (Tyson *et al*., 1972; Burton, Shuster & Cartlidge, 1975; McNeilly, 1975; Biswas & Rodeck, 1976; Sadovsky *et al*., 1977). Interestingly, these fluctuations in PRL correlate with similar fluctuations in sebum secretion seen throughout pregnancy and the post-natal period (Table 1). Furthermore, while parturition is associated with a normalisation of sebum secretion, in women that go on to breastfeed their newborns, sebum excretion rate remains elevated (Burton et al., 1970; Shuster & Thody, 1974; Burton, Shuster & Cartlidge, 1975; Henderson, Taylor & Cunliffe, 2000).

These observations highlight clear associations between sebum production, lactation, and the key regulatory hormone of lactation, PRL. Interestingly, the scent of areolar SG secretions from lactating women elicits autonomic and behavioural responses from neonates (Doucet *et al*., 2009). Therefore, one speculation as to the function of this putative PRL–SG axis is that it serves to facilitate maternal–neonatal chemocommunication (Nicholson, 1984).

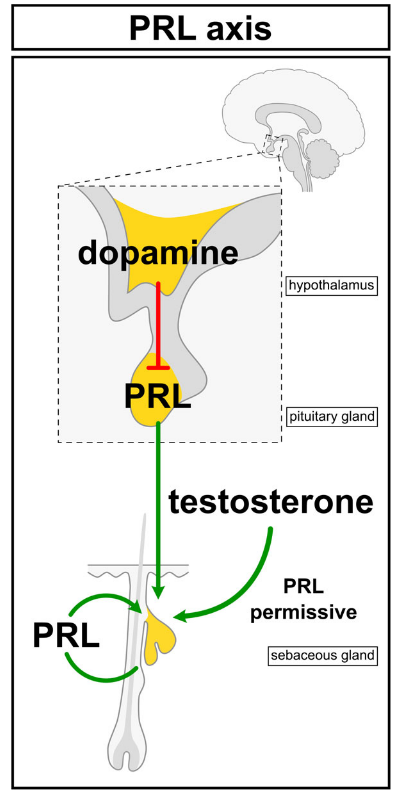

Fig. 7. Control of sebaceous glands (SGs) by the prolactin (PRL) axis. The secretion of PRL from lactotropic cells in the anterior pituitary is principally regulated by dopamine, which is secreted by tuberoinfundibular neurons in the hypothalamus, acting to inhibit PRL secretion. Disruption of this pathway in hypo-dopaminergic diseases, such as Parkinson's disease, results in excess levels of both PRL and sebum production. Pituitary-derived PRL may directly stimulate SGs and may also have a permissive role in stimulation of SGs by testosterone and other androgens. Intracutaneous sources of PRL could also act as paracrine hormones regulating SG function (see Fig. 6).

(2) Seborrhoea and prolactin in Parkinson's disease and phenylketonuria

Increased sebum secretion can also be seen in diseases characterised by abnormally high levels of serum PRL (Table 1). For example, in addition to the classic symptoms of tremor and bradykinesia, patients with Parkinson's disease (PD) exhibit elevated PRL levels (Agnoli *et al*., 1981; Wiesen, Yahr & Krieger, 1982; Winkler, Landau & Chaudhuri, 2002; Nitkowska *et al*., 2015) and excessive sebum production (Pochi, Strauss & Mescon, 1962a; Burton *et al*., 1973; Binder & Jonelis, 1983; Villares & Carlini, 1989; Martignoni *et al*., 1997). Under normal conditions, the release of PRL from the lactotropic cells of the anterior pituitary is inhibited by a tract of neurons referred to as the tuberoinfundibular pathway (Fitzgerald & Dinan, 2008) (Fig. 7).

Tuberoinfundibular neurons chiefly use dopamine as a neurotransmitter, and PD is the archetypical disease of dopaminergic neurons (Franceschi *et al*., 1988; Fitzgerald & Dinan, 2008). Both the seborrhoea and elevated PRL levels in PD can be treated by restoring normal levels of dopamine, e.g. by L-DOPA administration, or use of dopaminergic agonists (Burton, Cartlidge & Shuster, 1973; Kohn *et al*., 1973; Wiesen, Yahr & Krieger, 1982; Villares & Carlini, 1989; Winkler, Landau & Chaudhuri, 2002). These observations, combined with the positive correlations of PRL and sebum during pregnancy and lactation described above, strongly suggest a stimulatory role of PRL in the regulation of SGs and of sebum production (Franceschi *et al*., 1988; Fitzgerald & Dinan, 2008) (Fig. 7).

Phenylketonuria (PKU) is another hypo-dopaminergic condition which features both seborrhoea and elevated PRL levels (Table 1) (Burton, Goolamali & Shuster, 1975; Schulpis *et al*., 1998; Denecke *et al*., 2000; van Spronsen, 2010; Juhász *et al*., 2016). PKU is an autosomal recessive disease caused by inactivating mutations in the gene coding for phenylalanine hydroxylase, the enzyme responsible for tyrosine synthesis. Consequently, levels of tyrosine and downstream catecholamines, including dopamine, are reduced in those with PKU (van Spronsen, 2010). As with PD, the seborrhoea of PKU can be ameliorated by L-DOPA (Burton, Goolamali & Shuster, 1975). Overall, loss of dopamine-mediated inhibition of PRL secretion results in increased sebum production in PKU and PD.

Finally, anti-psychotic medication is another key factor, well recognised for its ability to induce hyperprolactinaemia, which must be carefully monitored before and during treatment (Bo *et al.*, 2016; González-Blanco *et al.*, 2016; Delgado-Alvarado *et al.*, 2018). Clinically, the resultant hyperprolactinaemia is associated with gynaecomastia, galactorrhoea, menstrual dysfunction, and, in the context of skin, with the development of hirsutism and acne (Garnis-Jones, 1996; Seeman, 2011; Bo *et al.*, 2016).

(3) Mechanisms of prolactin-mediated regulation of sebaceous glands

PRL partly restores sebum production in hypophysectomised–castrated rats when administered alongside testosterone (Ebling, Ebling & Skinner, 1969a). However, PRL has a limited effect on stimulating sebum production by itself in these animals (Ebling, Ebling & Skinner, 1969a; Nikkari & Valavaara, 1970) (Fig. 4). These observations suggest that, in the context of sebum production, the main function of PRL might be to sensitise SGs to stimulation by androgens (Fig. 7), perhaps by upregulating expression of androgen-metabolising enzymes (Langan, Hinde & Paus, 2018). Indeed, PRL is known to regulate the activity of 5α-reductase peripherally, namely in the reproductive organs, but in this context, PRL appears to inhibit 5α-reductase (Yamanaka *et al.*, 1975; Noci *et al.*, 1986; Serafini & Lobo, 1986). Furthermore, inhibition of 5α-reductase *via* dutasteride or finasteride has no effect on sebum production (Imperato-McGinley *et al.*, 1993; Bhasin *et al.*, 2012), suggesting that modulation of androgen metabolism might not be a crucial component of PRL-mediated regulation of SGs.

It also remains a possibility that PRL acts directly on human SGs, despite the observed lack of effect of PRL on sebum production in rodents (Ebling, Ebling & Skinner, 1969a; Nikkari & Valavaara, 1970). The PRL receptor (PRL-R) has been detected by immuno-histochemistry in the cutaneous SGs of mice and sheep, and in SGs associated with the mouse nipple (Choy, Nixon & Pearson, 1997; Craven *et al.*, 2001; Koizumi *et al.*, 2003). Human SGs also reportedly demonstrate PRL-R immuno-reactivity (Foitzik, Langan & Paus, 2009). It is known that mouse and human HFs are both a source and target of PRL in human skin (Foitzik et al., 2003, 2006; Foitzik, Langan & Paus, 2009; Paus *et al.*, 2014) (Fig. 6), which serves as proof-of-principle that PRL can act directly on cutaneous appendages. Furthermore, a recent study, utilising full-thickness human skin cultured *ex vivo*, has provided preliminary evidence that PRL increases the size and lipid content of SGs, and also stimulates proliferation of peripheral sebocytes (Langan, Hinde & Paus, 2018). These effects of PRL could be ameliorated using a mutant form of human PRL (Δ1–9-G129R-hPRL) as a competitive PRL-R antagonist (Langan, Hinde & Paus, 2018).

Of course, it is difficult to determine the extent to which circulating PRL is the principal driver of changes in SG physiology in diseases that feature abnormal PRL levels, as these often also involve a number of other endocrine axes (Karaca *et al.*, 2010; Nitkowska *et al.*, 2015). This is further complicated by the fact that PRL-R can be bound and activated by other hormonal ligands, such as GH (Xu *et al.*, 2013). Compounding this problem, both PRL and PRL-R are expressed in both healthy and diseased human skin (Foitzik *et al.*, 2006; El-Khateeb *et al.*, 2011; Yang *et al.*, 2018), suggesting that local intra-cutaneous PRL production may also regulate SG function (Fig. 6).

The relative importance of this putative PRL–SG axis to each gender is also unclear. If the primary physiological function of the PRL–SG axis is to increase perinatal sebum production, then it is perhaps surprising that the same mechanism is evidently present in men (Cotterill *et al.*, 1971; Burton, Goolamali & Shuster, 1975; Schulpis *et al.*, 1998; Nitkowska *et al.*, 2015). A reasonable hypothesis is that the PRL-SG axis in males somehow only becomes conspicuous when systemic levels of PRL are pathologically high.

Experimentally, the use of short-term *ex vivo* SG culture systems could be readily exploited to dissect the contribution of intra-cutaneous PRL production *versus* systemic PRL in the regulation of SG physiology, as well as to identify any sex-dependent effects (Langan, 2018; Langan, Hinde & Paus, 2018). This would remove the influence of ill-defined serum-derived endocrine factors, thus allowing for the drawing of more definitive conclusions regarding the role of PRL. Indeed, the same approach could be used to investigate the roles of any intra-cutaneous or sebocyte-derived neurohormone.

VI. GROWTH HORMONE AND INSULIN-LIKE GROWTH FACTOR

The peptide hormone GH is produced chiefly by the somatotropic cells of the anterior pituitary (Fig. 8). Excessive GH production in adulthood results in a condition known as acromegaly: manifesting as enlargement of the hands, feet, and parts of the head, including the supraorbital ridges, nose, jaw, and tongue. Acromegaly and GH deficiency are also associated with a number of skin changes (Lange *et al.*, 2001; Kanaka-Gantenbein *et al.*, 2016). Both acromegalic men and women exhibit greatly increased sebum secretion (Burton *et al.*, 1972; Borlu *et al.*, 2012; Paus, 2016*b*) (Table 1). Conversely, GH deficiency is associated with abnormally low sebum secretion, which can be normalised using GH-replacement therapy (Goolamali, Burton & Shuster, 1973; Pochi & Strauss, 1974; Tanriverdi *et al.*, 2006; Borlu *et al.*, 2007).

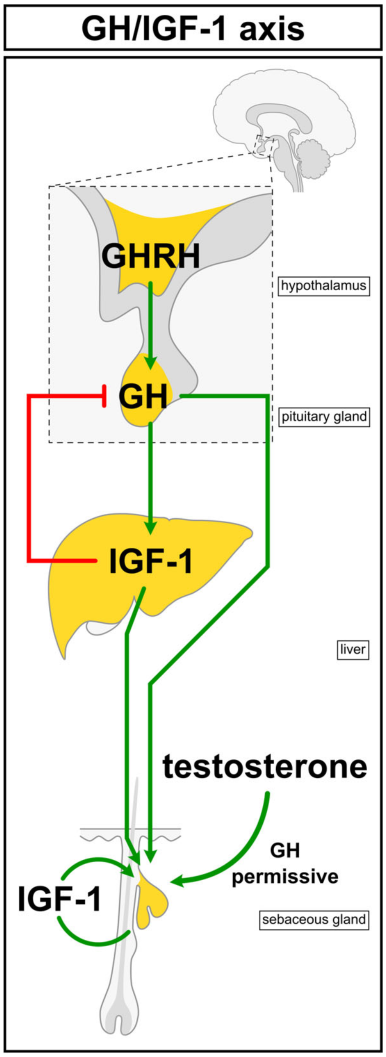

Fig. 8. Control of sebaceous glands (SGs) by the growth hormone (GH)/insulin-like growth factor-1 (IGF-1) axis. Hypothalamic growth hormone-releasing hormone (GHRH) elicits release of growth hormone (GH) from somatotropic cells in the anterior pituitary. GH promotes IGF-1 production in hepatocytes, which in turn stimulates SGs and forms a negative feedback loop to inhibit GH secretion. GH may act directly on SGs and also facilitates the stimulatory action of testosterone on SGs. Sebocyte-derived IGF-1 might also function as an autocrine hormone (see Fig. 6).

In hypophysectomised–castrated rats, GH restores SG sensitivity to testosterone, but has a limited effect on its own (Ebling, Ebling & Skinner, 1969a; Ebling *et al*., 1975a) (Fig. 4). However, in cultures of rat preputial sebocytes, physiological concentrations of GH cause a significant increase in the number of lipid-producing preputial cells (Deplewski & Rosenfield, 1999). This finding has been corroborated in SZ95 sebocytes (Makrantonaki *et al*., 2008) suggesting that GH may directly affect sebocytes. Indeed, both rodent and human SGs have demonstrated immuno-reactivity for the GH receptor (Lobie *et al*., 1990; Oakes *et al*., 1992; Kővári *et al*., 2018).

Acromegaly and GH deficiency are further characterised by respectively increased, and decreased, levels of insulin-like growth factor 1 (IGF-1), which is a major downstream hormonal mediator of GH (Kwan & Hartman, 2007; Oldfield *et al*., 2017) (Fig. 8). IGF-1 and insulin signalling are mediated through two receptor tyrosine kinases, the insulin receptor and IGF-1 receptor (IGF-1R), which exhibit partial redundancy through ligand cross-reactivity (Boucher, Tseng & Kahn, 2010; Jung & Suh, 2015). SGs *in vivo* demonstrate immuno-reactivity for IGF-1R (Hodak *et al*., 1996; Rudman *et al*., 1997), and treatment of SZ95 and SEB-1 immortalised sebocytes with IGF-1 results in increased lipogenesis (Smith *et al*., 2006; Makrantonaki *et al*., 2008; Im *et al*., 2012) and increased expression of inflammatory mediators (Im *et al*., 2012). Similar observations of IGF-1 mediated stimulation of lipid production have been made using primary human sebocytes, which also express IGF-1R (Hwang *et al*., 2016).

The lipogenic effects of IGF-1 are mediated by increasing expression of sterol regulatory element-binding proteins (SREBPs) *via* activation of the phosphoinositide 3-kinase/serine-threonine protein kinase AKT1/mammalian target of rapamycin (PI3K/Akt/mTOR) pathway (Smith *et al*., 2006; Im *et al*., 2012; Mirdamadi *et al*., 2015; Hwang *et al*., 2016). In turn, PI3K/Akt/mTOR activation results in inhibitory phosphorylation of forkhead box (FoxO) transcription factors (Mirdamadi *et al*., 2015) which, in the context of adipogenesis, inhibit PPARγ-mediated gene transcription (Fan *et al*., 2009). In line with this model, isotretinoin treatment in acne patients reportedly results in nuclear translocation of non-phosphorylated FoxO1 and FoxO3 in SGs (Agamia *et al*., 2018).

The PI3K/Akt/mTOR pathway is also known to stimulate proliferation in other tissues (Yu & Cui, 2016). As expected, IGF-1 increases both proliferation and the number of lipid-producing cells in rat preputial sebocytes (Rosenfield *et al*., 1998). It should be noted however that there are conflicting data on the ability of IGF-1 to stimulate proliferation of SZ95 cells (Im *et al*., 2012; Mirdamadi *et al*., 2015). Overall, GH may have the capacity to stimulate sebocytes directly, but likely exerts most of its effects on the SG indirectly through IGF-1 (Fig. 8). Both GH and IGF-1 stimulate lipogenesis, but so far only IGF-1 has been shown to promote cell division in sebocytes (Rosenfield *et al*., 1998; Mirdamadi *et al*., 2015). Furthermore, SGs *in vivo* are immuno-reactive for IGF-1 (Ghahary *et al*., 1998), and transcribe mRNA for the IGF-1 binding proteins, IGFBP2 and IGFBP4 (Batch *et al*., 1994). These findings suggest that IGF-1 may promote lipid synthesis and proliferation in sebocytes as an autocrine factor (Fig. 6). Such an autocrine mechanism could be facilitated by pituitary-derived GH, similar to how GH induces IGF-1 production in hepatocytes (Takahashi, 2017) (Fig. 8). While mRNA for GH has been detected in human dermis (Slominski *et al*., 2000*a*), it is not known whether human skin synthesises any appreciable quantity of GH peptide.

The evident involvement of IGF-1 in the regulation of sebocyte biology is especially interesting given the known role of insulin/IGF-1 signalling in mediating the systemic response to glucose, and the association of reduced acne severity with consumption of low glycaemic-load diets (Smith *et al.*, 2007, 2008; Kwon *et al.*, 2012). Levels of IGF-1 have been found to correlate positively with sebum production, and are increased in people with acne (Aizawa & Niimura, 1995; Cappel, Mauger & Thiboutot, 2005; Vora *et al.*, 2008; Ji *et al.*, 2018). Furthermore, congenital IGF-1 deficiency is protective against acne development (Ben-Amitai & Laron, 2011). Taken together, these observations strongly suggest that IGF-1 signalling plays a role in acne pathogenesis. Levels of GH may therefore affect predisposition to acne by modulating insulin/IGF-1 signalling.

VII. HYPOTHALAMIC–PITUITARY–THRYOID AXIS

The thyroid gland regulates metabolism *via* the production of thyroid hormones, triiodothyronine (T_3) and thyroxine (T_4). In peripheral tissues, T_4 is converted to the more potent T_3 (Fig. 9). Thyrotropin-releasing hormone (TRH), produced by the hypothalamus, stimulates the thyrotropic cells of the anterior pituitary to release thyrotropin (TSH or thyroid-stimulating hormone). TSH, in turn, elicits release of thyroid hormones from the thyroid gland, which act on tissues throughout the body (Zoeller, Tan & Tyl, 2007) (Fig. 9).

Hypothyroidism is associated with abnormally low sebum secretion (Goolamali, Evered & Shuster, 1976; Paus, 2016*b*) (Table 1). Similarly, surgical removal of the thyroid gland in rats results in reduced sebum production (Thody & Shuster, 1970b, 1972). L-thyroxine can partly restore normal sebum secretion in hypothyroid patients (Goolamali, Evered & Shuster, 1976). Conversely, thyrotoxicosis (hyperthyroidism) is associated with increased sebum secretion, but reportedly only in women (Goolamali, Evered & Shuster, 1976). This may perhaps reflect that, physiologically, male SGs are already maximally stimulated by other hormones, such as androgens (Pochi & Strauss, 1974).

TSH and T_4 increase sebum secretion when administered to rats, and can partly counteract the reduction in sebum production due to castration and hypophysectomy (Ebling, Ebling & Skinner, 1970; Thody & Shuster, 1972) (Fig. 4). Importantly, TSH treatment in thyroidectomised animals does not produce a sebaceous response, indicating that TSH-mediated stimulation of sebum secretion depends on the presence of the thyroid gland (Thody & Shuster, 1972).

Thyroid hormones seemingly act on the SG independently of androgens, given that TSH and T_4 increase sebum secretion even in castrated animals, and that combined treatment of castrated animals with T_4 and testosterone does not appear to be synergistic (Ebling, Ebling & Skinner, 1970; Thody & Shuster, 1972; Toh, 1981). Additionally, T_3 was shown to almost double the lipogenesis rate in *ex vivo* cultured human SGs (Guy, Ridden & Kealey, 1996). In further support of a direct action of thyroid hormones on the SG (Fig. 9), the presence of the α1-thyroid hormone receptor has been demonstrated in human scalp SGs (Ahsan *et al.*, 1998), while the CTS surrounding human SGs has also been reported to be immuno-reactive for the TSH receptor (Bodó *et al.*, 2009). The TSH receptor, along with deiodinases 2/3 and TSHβ, have also been demonstrated in human keratinocytes and whole-skin biopsies (Slominski *et al.*, 2002).

It is currently unknown whether the SG synthesises TSH or TRH, or is capable of converting T_4 to T_3. The HF, however,

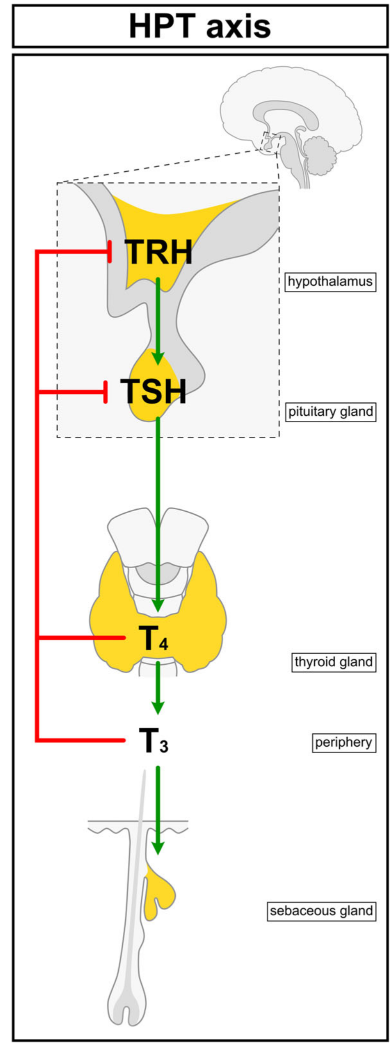

Fig. 9. Control of sebaceous glands (SGs) by the hypothalamic–pituitary–thyroid (HPT) axis. Hypothalamic thyrotropin-releasing hormone (TRH) stimulates secretion of thyroid-stimulating hormone (TSH) from thyrotropic cells in the anterior pituitary. In turn, TSH acts on the thyroid gland, stimulating secretion of T_4, which is converted to active T_3 in peripheral tissues. T_3 then stimulates sebum production. Thyroid hormones participate in a negative feedback loop that inhibits TSH and TRH secretion.

is known to produce TRH and express TRH-R within the outer root sheath (Gáspár *et al.*, 2010). Both TRH and TSH, as well as thyroid hormones, increase biogenesis and activity of mitochondria when administered to human HFs *ex vivo* (Vidali *et al.*, 2014), raising the question of whether similar mitochondrial effects are also seen in the SG. While the observed data regarding TSH administration to thyroidectomised rats would suggest that SGs do not respond to HPT-axis neurohormones directly (Ebling, Ebling & Skinner, 1970; Thody & Shuster, 1972), this has yet to be explicitly tested in human SGs. However, the reported observation of TSH receptor protein in the mesenchyme of both human HFs and SGs (Bodó *et al.*, 2009), tentatively suggests that TSH exerts some direct effect on human SGs.

VIII. SEBACEOUS NEUROBIOLOGY

(1) Clinical evidence for regulation of sebaceous glands by the peripheral nervous system

The skin is a densely innervated organ. Neuropeptides and neurotransmitters released from cutaneous nerve fibres control many important processes in the skin, such as epidermal proliferation and apoptosis, wound healing, inflammation, and melanogenesis (Hara *et al.*, 1996; Harvima, Nilsson & Naukkarinen, 2010; Lebonvallet *et al.*, 2012; Blais *et al.*, 2014; Cheret *et al.*, 2014). Furthermore, most skin appendages have a well-described nerve supply, such as the eccrine gland and HF (Montagna & Parakkal, 1974a, 1974c; Botchkarev *et al.*, 1997, 1999; Schulze *et al.*, 1997). However, it has not yet been conclusively demonstrated whether SGs are also directly innervated.

The existence of neuronal control over SG function is, however, indicated by both historical and more recent clinical observations (Table 3). In patients with partial facial paralysis, variations of sebum secretion rate can be observed between the normal and impaired halves of the face (Burton *et al.*, 1971; Summerly, Woodbury & Boddie, 1971). Unilateral facial acne has also been observed following partial facial paralysis (Tagami, 1983; Sudy & Urbina, 2018). Sebum secretion from the thighs of paraplegic patients is significantly higher than that in controls, while the forehead sebum excretion rate is unchanged (Thomas *et al.*, 1985). Furthermore, a significant decrease in sebum secretion can be achieved using topical anti-muscarinic agents (Cartlidge, Burton & Shuster, 1972), and botulinum toxin-A ('Botox') has been used to reduce acne severity and sebum secretion in humans (Jankovic & Diamond, 2006; Shah, 2008; Li *et al.*, 2013b; Min *et al.*, 2015). Together, these observations suggest that functional neurotransmission is important for maintaining normal levels of sebum secretion (Table 3).

(2) Characterising sebaceous gland innervation

Previous studies that have attempted to demonstrate the putative sebaceous nerve supply relied primarily on the use of histochemical stains, such as silver nitrate deposition, or enzymatic staining using modified cholinesterase substrates (Champy, Coujard & Coujard-Champy, 1945; Winkelmann & Schmit, 1959; Hashimoto, Ogawa & Lever, 1963; Pawlowski & Weddell, 1967). No study has yet been able to accurately map the extent of sebaceous nerve fibres, nor determine where these nerve fibres terminate. The cholinesterase technique in particular has produced conflicting results, with authors making contrary claims as to the presence of acetylcholine (ACh)-secreting nerve fibres in proximity to the periphery of the SG (Hurley, Shelley & Koelle, 1953; Hellmann, 1955; Montagna & Ellis, 1957). However, limited ultrastructural evidence of the presence of nerves in the SG connective tissue has been reported in rat (Pawlowski & Weddell, 1967; Dugan, 1974) and camel skin (Taha, 1988) (Table 3).

The most recent report regarding sebaceous innervation described substance P (SP)-positive nerve fibres in the stroma of human SGs, yet only in acne lesion-associated glands (Toyoda *et al.*, 2002). In addition, the reduction in sebum secretion that has been achieved using botulinum toxin-A injections occurred only in patients who had seborrhoea prior to treatment (Li *et al.*, 2013b). Taken together, these findings paint a somewhat confusing image of SG innervation and raise the possibility that nervous input to human SGs might only exist, or become manifest, in the context of SG dysfunction. This may explain the historical difficulty in identifying and accurately mapping sebaceous neuroanatomy.

From research on Meibomian glands, it appears that sebaceous innervation does indeed occur *in vivo*, albeit in this specialised form of SG. Meibomian glands are embedded within the tarsal plate of the eyelid, and therefore are also known as tarsal glands (Butovich, 2017). A substance comparable to sebum ('meibum') is derived from meibocytes, similar to how sebum is generated by sebocytes. Meibum is thought to reduce evaporation of the tear film that covers the eye (Butovich, 2017). Meibomian glands exhibit an extensive network of nerve fibres within their surrounding connective tissue, and these nerves produce several key neurotransmitters and neuropeptides, including calcitonin gene-related peptide, vasoactive intestinal peptide, SP and ACh (Montagna & Parakkal, 1974*b*; Seifert & Spitznas, 1996; Cox & Nichols, 2014). At the ultrastructural level, unmyelinated axons can be seen to pass within 1 μm of the cell membranes of basal meibocytes (Seifert & Spitznas, 1996).

A further, as yet unexplored, possibility is that nerve fibres control the SG not by direct innervation, but by acting on the known sebaceous progenitors within the HF (Füllgrabe *et al.*, 2015; Barnes, Saurat & Kaya, 2017) (Fig. 10). The HF is supplied by a well-characterised neuronal network, consisting of a dense accumulation of perifollicular and longitudinal nerve endings that associate closely with follicular keratinocytes at the level of the upper bulge/lower JZ, and a more loose collection of nerve endings serving the infundibulum (Montagna & Parakkal,

Table 3. Evidence for neurobiological regulation of sebaceous gland (SG) function

Implicated neurotransmitter, neuropeptide, neurotrophin, or receptor	Species/system	Methodology	Findings	Reference(s)
n/a	Rat and camel SGs	TEM	Reported basal and intraepidermal nerve fibres	Pawlowski & Weddel (1967); Dugan (1974); Taha (1988)
n/a	Human patients with facial paresis	Measurement of sebum excretion rate	Dysregulation of sebum excretion on affected side	Burton *et al.* (1971); Summerly, Woodbury & Boddie (1971)
n/a	Human paraplegic patients	Measurement of sebum excretion rate	Increase in sebum excretion below neurological lesion	Thomas *et al.* (1985)
n/a	Human *in vivo*	Intradermal botulinum toxin injection	Reduction in sebum excretion	Jankovic & Diamond (2006); Shah (2008); Z.J. Li *et al.* (2013b); Min *et al.* (2015)
ACh	Human SGs	Enzyme histochemistry	Cholinesterase activity in nerve fibres around SGs	Montagna & Ellis (1957)
ACh	Human *in vivo*	Topical treatment with anti-cholinergic (poldine methyl-methosulphate)	Reduction in sebum production	Cartlidge, Burton & Shuster (1972)
ACh	Primary human facial sebocytes	Incubation with ACh	Increase in lipid droplet formation	Li *et al.* (2013b)
nAChRα7		Incubation with α-bungarotoxin	Blockade of ACh-mediated lipid synthesis	
nAChRα7		WB, PCR	nAChRα7 expression	
nAChRα7	Mouse	IHC	nAChRα7 in SGs	Fan *et al.* (2011)
α3, α7, α9, m2, m3, m4, m5, β2 & β4 ACh receptors	Human SGs	IF, IHC	ACh receptor expression	Kurzen *et al.* (2004); Z.J. Li *et al.* (2013b)
SP	Rat SGs	IHC	SP-reactive nerve fibres around SGs	Ruocco *et al.* (2001)
SP	Human SGs	IHC	SP-reactive nerves around acne-lesional SGs	Toyoda *et al.* (2002)
SP	Human skin *ex vivo*	Incubation with SP	Increase in SG and sebocyte cross-sectional area	Toyoda & Morohasi (2003)
SP	SZ95 sebocytes	Incubation with SP	Increased expression of IL-1, IL-6, TNFα and PPARγ	Lee *et al.* (2008)
NT3, NGF, GDNF, TrkA, TrkC, p75, GFRα1/2	Human SGs	IF, IHC	Expression of neurotrophins and cognate receptors	Toyoda, Nakamura & Morohashi (2002); Adly *et al.* (2005, 2006a, 2006b); Adly, Assaf & Hussein (2009)

1974*c*; Botchkarev *et al.*, 1997, 1999, 2004; Peters *et al.*, 2001, 2002; Li & Ginty, 2014; Cheng *et al.*, 2018) (Fig. 10). The number of nerve fibres comprising these follicular neuronal networks, as well as the presence of peptidergic and non-peptidergic neurotransmitters, exhibits profound hair-cycle dependence (Botchkarev *et al.*, 1997, 1999; Peters *et al.*, 2001). It can be hypothesised that the known sebaceous progenitors within the HF are subject to regulation by these neurons, or are at least reliant on the presence of peri-neuronal extracellular matrix components (Cheng *et al.*, 2018). Future research must aim to correlate regions of dense pilosebaceous innervation with the presence of SG progenitor populations throughout the hair growth cycle.

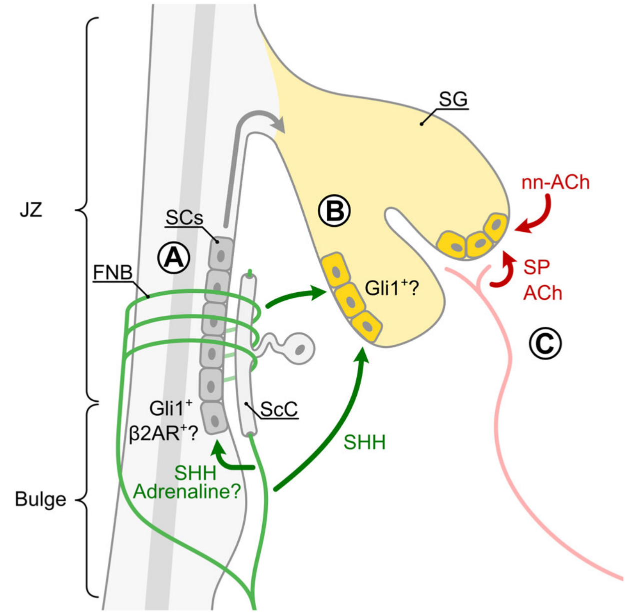

Fig. 10. Regulation of sebaceous glands (SGs) by cutaneous nerves and neuromediators: working hypotheses based on available evidence. (A) Neuronal regulation of SG stem cells (SCs) in the hair follicle (HF) epithelium. The junctional zone (JZ) contains leucine-rich repeats and immunoglobulin-like domains protein 1-expressing (Lrig1+) cells and leucine-rich repeat-containing G-protein coupled receptor 6-expressing (Lgr6+) cells, which represent potential SG progenitor populations. The isthmus (JZ and bulge) region of the HF is also densely innervated (known as follicular nerve network B; FNB), and Lgr6 expression is dependent on intact innervation and the presence of Schwann cells (ScC). Zinc finger protein GLI1 (Gli1) expression in the upper bulge is dependent on nerve-derived sonic hedgehog (SHH) and it is unknown if these Gli1+ cells might also contribute to the SG under certain conditions. Β2-adrenoreceptor (β2AR) expression has also been reported in the isthmus/bulge region of the HF. (B) Nerve-derived SHH might act on Gli1+ sebocytes reported in the peripheral zone (PZ). (C) Direct regulation of peripheral sebocytes by other neuromediators including substance P (SP), neuronal acetylcholine (ACh) from dermal nerve fibres, and non-neuronal sources of ACh (nn-ACh) such as keratinocytes.

(3) Effects of cutaneous denervation on murine sebaceous glands

Denervation of mouse skin has been a valuable experimental technique for investigating the role of nerve fibres in skin homeostasis and disease. While there are currently no reports of SG phenotypes following experimental denervation, certain findings have interesting implications, drawing on what is known about SG homeostasis. For example, expression of Lgr6 in the PSU and epidermis has been shown to depend on intact innervation (Liao & Nguyen, 2014). As previously discussed, Lgr6-expressing cells are long-lived progenitor cells, and intra-sebaceous Lgr6+ cells contribute to the murine SG (Gong *et al.*, 2012; Page *et al.*, 2013; Liao & Nguyen, 2014; Kretzschmar *et al.*, 2016). Schwann cells are closely associated with Lgr6-expressing cells in the mouse HF and epidermis, and denervation of mouse dorsal skin leads to a loss of both Schwann cells and Lgr6 expression in both compartments (Liao & Nguyen, 2014). Additionally, Lgr6+ cells throughout the PSU are enriched for genes related to axonal outgrowth and synapse maintenance, which could be indicative of direct innervation of these cells (Füllgrabe *et al.*, 2015).

Another finding of potential interest is that in mouse skin, sonic hedgehog (Shh) is produced by a sub-population of follicular nerves, which is required for maintenance of zinc finger protein GLI1 (Gli1) expression in cells of the upper bulge (Fig. 10). Lineage tracing demonstrates that these cells not only contribute to the cycling portion of the HF, but can also regenerate the epidermis and contribute to the SG following wounding (Brownell *et al.*, 2012). RNA sequencing of isolated Gli1+ cells from mouse skin has revealed that these cells express several genes related to neurogenesis, suggestive of direct innervation (Cheng *et al.*, 2018). Moreover, given the long-range signalling capabilities of Shh (Goetz *et al.*, 2002), it is conceivable that nerve-derived Shh also acts on Gli1-expressing cells that have been reported within the SG (Niemann *et al.*, 2003) (Fig. 10).

From the work described above, it can be hypothesised that denervation may result in SG atrophy due to loss of progenitor cells and trophic factors, such as Shh. Future research must systematically interrogate the effects of cutaneous denervation on SG biology and sebum production. If nerves serve only a limited role in SG homeostasis, but can influence SG function in response to various stressors or under pathological conditions, then researchers may aim to identify which conditions and stimuli elicit the development of neuro-regulation, and compare the ability of such stimuli to generate SG phenotypes in intact and denervated backgrounds.

(4) Acetylcholine

All branches of the nervous system extensively use ACh as an excitatory neurotransmitter. Neurons that synthesise ACh can be detected using colorimetric staining techniques that rely on the activity of cholinesterase, an enzyme involved in ACh metabolism (Saxod, 1978). As previously stated, cholinesterase-positive nerve fibres have been observed within the CTS of human SGs (Table 3) (Montagna & Ellis, 1957). Expression of cholinergic receptors has also been demonstrated in human SGs. For example, the muscarinic receptor m2AChR is reportedly detectable in suprabasal sebocytes *via* immuno-fluorescence, while the nicotinic receptors nAChRα7, nAChRα10 and nAChRβ4 are expressed throughout the basal, suprabasal and ductal cells of the SG (Kurzen *et al.*, 2004). Of particular note is nAChRα7, of which both mRNA and protein is produced in human primary sebocytes (Kurzen *et al.*, 2004; Li *et al.*, 2013b). Mouse SGs are also immuno-reactive for nAChRα7 (Fan *et al.*, 2011).

Treatment of primary human facial sebocytes with ACh dose-dependently increases lipid production (1–10 nM ACh), and this effect can be partially abrogated using the nAChRα7 antagonist, alpha-bungarotoxin (Li *et al.*, 2013b) (Table 3). These observations suggest that sebocytes can be directly stimulated by nerve-derived ACh, however, stimulation by non-neuronal sources of ACh in the skin remains a possibility (Wessler *et al.*, 2003; Kurzen *et al.*, 2007; Wessler & Kirkpatrick, 2008) (Fig. 10). In further support of a direct action of ACh on SGs, treatment of immortalised human meibocytes with the broad-spectrum cholinergic agonist carbachol causes an increase in proliferation and an influx of intracellular calcium (Kam & Sullivan, 2011). Furthermore, concentrations of pilocarpine akin to those used in anti-glaucoma eye drops are able to inhibit epidermal growth factor (EGF)- and bovine pituitary extract-mediated proliferation of immortalised human meibocytes (Zhang *et al.*, 2017).

(5) Adrenaline

Adrenaline (epinephrine) is a major neurotransmitter of the sympathetic nervous system and a hormonal product of the adrenal glands. Propranolol-mediated blockade of adrenergic β-receptors over 8–10 weeks does not affect either sebum excretion rate or acne grade in patients with acne vulgaris (Cunliffe & Cotterill, 1970). This is consistent with earlier observations that intra-dermal injections of adrenaline also do not alter the rate of sebum secretion in humans (Miescher & Schonberg, 1944; Ikai, 1958; Kligman & Shelley, 1958). Furthermore, mapping of adrenergic receptor expression in skin *via* the use of radio-labelled ligands reveals that SGs do not express either β1- or β2-adrenergic receptors (Steinkraus *et al.*, 1996).

However, the rat preputial gland is known to receive adrenergic innervation (Ambadkar & Vyas, 1981), and has been shown to increase secretion in response to adrenaline and phenylephrine, an α-adrenoceptor agonist (Ambadkar & Vyas, 1980). Blockade of α-adrenoceptors, but not β-adrenoceptors, inhibits preputial gland secretion (Ambadkar & Vyas, 1980). Differentiation of human Meibomian gland cells has also been shown to be positively regulated by brimonidine, an α2-adrenoreceptor agonist (Han *et al.*, 2018). Furthermore, in mice, lack of Meibomian gland innervation by dopamine beta hydroxylase-expressing neurons is associated with blepharoptosis (drooping eyelids) and corneal ulceration, which are symptoms of Meibomian gland dysfunction (Eldredge *et al.*, 2008). Therefore, it cannot yet be ruled out that adrenergic innervation and catecholamines play a role in SG physiology, at least through α-adrenoceptors, and in selected mammalian species and SG sub-types (Ambadkar & Vyas, 1980).

Interestingly, the β2-adrenocreceptor is selectively upregulated in the isthmus of murine HFs during early anagen (Botchkarev *et al.*, 1999). Given recent evidence indicating a hair-cycle dependent upregulation of SG size (Reichenbach *et al.*, 2018), it can be hypothesised that the abrupt and transient expression of the β2-adrenocreceptor reflects a similarly brief involvement of adrenergic signalling in recruitment of sebaceous progenitors to the SG (Fig. 10). This model might also explain why attempts to modulate sebum secretion in humans with adrenergic compounds have been unsuccessful (Miescher & Schonberg, 1944; Ikai, 1958; Kligman & Shelley, 1958; Cunliffe & Cotterill, 1970), given the small percentage of HFs undergoing telogen-to-anagen transition in human skin at any particular time (van Scott, Reinertson & Steinmuller, 1957; Kligman, 1959; Saitoh, Uzuka & Sakamoto, 1970).

(6) Substance P

SP is the primary member of the tachykinin family of neuropeptides. In skin, SP modulates nociception, inflammation, and vasodilation (Polak & Bloom, 1981; Paus, Theoharides & Arck, 2006; Sahbaie *et al.*, 2009; Wei *et al.*, 2012). Nerve fibres that produce this prototypic stress- and neurogenic inflammation-associated neuropeptide have been shown to pass within the CTS of rat SGs (Table 3) (Ruocco *et al.*, 2001). SP-immuno-reactive nerve fibres have also been observed in the connective tissue surrounding human SGs (Fig. 10), but only in glands that are associated with acne lesions (Toyoda *et al.*, 2002), as stated previously.

Treatment of mouse skin with 0.1μM SP *ex vivo* increases the cross-sectional area of SGs and individual sebocytes, as well as the number of vacuoles in differentiating sebocytes *in situ*, indicating heightened lipid production (Toyoda & Morohashi, 2003). Primary human sebocytes, treated with 1μM SP, exhibit a pro-inflammatory phenotype, indicated by increased production of interleukins IL-1 and IL-6, as well as increased expression of tumour necrosis factor-α (TNFα) and PPARγ (Lee *et al.*, 2008). These effects of SP can be partially reduced by co-application with the corticosteroid dexamethasone, which has well-described anti-inflammatory effects (Newton, 2000; Lee *et al.*, 2008). The evident pro-lipogenic and pro-inflammatory action of SP on sebocytes would implicate SP as a potential aetiologic factor within the proposed inflammatory model of acne pathogenesis (Tanghetti, 2013; Zouboulis, Jourdan & Picardo, 2014).

(7) Neurotrophins

Neurotrophins are a class of secreted proteins that guide axonal outgrowth and provide trophic support to neurons (Skaper, 2012). Immuno-reactivity for neurotrophin-3 (NT3) (Adly *et al.*, 2005), nerve growth factor (NGF) (Toyoda & Morohashi, 2003; Adly *et al.*, 2006*b*), glial-derived neurotrophic factor (GDNF) (Adly *et al.*, 2006*a*), and pigment epithelium-derived factor (PEDF) (Li *et al.*, 2013a) have all been reported in human SGs (Table 3). The presence of neurotrophins does not appear to be restricted to any particular zone of the SG, except for GDNF, which is confined to peripheral sebocytes (Adly *et al.*, 2006*a*). Furthermore, NGF protein can be detected in the peripheral sebocytes of acne-associated SGs, but not in the glands of healthy skin (Toyoda, Nakamura & Morohashi, 2002; Toyoda & Morohashi, 2003).

Interestingly, NGF expression can be induced in peripheral sebocytes by incubation of human skin with IL-6 (Toyoda & Morohashi, 2003). Sebaceous production of neurotrophins supports the hypothesis that the SG epithelium can attract a supply of nerve fibres to the gland, possibly under the pro-inflammatory conditions associated with acne. Human SGs also exhibit immuno-reactivity for several neurotrophin receptors including the tropomyosin receptor kinases TrkA and TrkC, the GDNF receptor GFRα1/2 and the low-affinity nerve growth factor receptor (p75NTR) (Adly *et al.*, 2005, 2006*b*,*a*; Adly, Assaf & Hussein, 2009), suggesting that neurotrophins can exert a direct effect on sebocytes.

Yet, what role neurotrophins may play in the control of sebocyte function remains unclear. A striking historical report demonstrated that subcutaneous injection of murine NGF (0.2 mg/ml) in mice produces a threefold enlargement of SGs near the injection site after two weeks (Reams, 1967). Furthermore, mice that are null for Egr3 (early growth response 3), a downstream effector of NGF signalling, develop symptoms of Meibomian gland dysfunction along with loss of Meibomian gland innervation (Eldredge *et al.*, 2008). A distinction must be made here between several different potential modes of action of neurotrophins on SGs. The reported expression of both neurotrophins and neurotrophin receptors would suggest the existence of a paracrine, or possibly autocrine, mechanism of neurotrophin regulation of sebocytes (Table 3) (Toyoda & Morohashi, 2003; Adly *et al.*, 2005, 2006*b*,2006*a*, Adly, Assaf & Hussein, 2009). Alternatively, neurotrophins may indirectly control SG function by modulating the recruitment of nerve fibres to either the SG or perhaps to sebaceous progenitor cells within the follicular epithelium.

Interestingly, HF morphogenesis and cycling in mice depends on neurotrophins (Botchkarev *et al.*, 2004; Peters *et al.*, 2006; Zhou *et al.*, 2006), although cutaneous denervation only moderately affects hair growth and hair cycle progression under wounding conditions (Maurer *et al.*, 1998; Brownell *et al.*, 2012). This would suggest that neurotrophins primarily act as paracrine and/or autocrine hormones within the PSU, rather than solely promoting innervation, although it remains to be seen whether this is true for the SG specifically.

(8) Endocannabinoids

Endocannabinoids (ECs) are a group of neuro-modulatory, poly-unsaturated fatty acids that are synthesised throughout the mammalian nervous system, but also in many other tissues, including cutaneous epithelia (Maccarrone *et al.*, 2015). In the nervous system, ECs, such as anandamide (AEA) or 2-arachidonoylglycerol (2-AG), are synthesised and released from the post-synapse in response to depolarisation of the post-synaptic membrane. These ECs then bind and activate G-protein-coupled receptors within the pre-synaptic membrane. Activation of these cannabinoid receptors (CB1 or CB2) has several effects including modulation of neuronal excitability and altering gene expression. Cannabinoids can also bind and activate transient receptor potential (TRP) receptors (Caterina, 2014).

Plant-derived cannabinoid compounds are referred to as phytocannabinoids, and these include the psychoactive tetrahydrocannabinol (THC) from the *Cannabis* plant (Di Marzo & Piscitelli, 2015). Collectively, ECs, their associated receptors, and the enzymes involved in EC synthesis throughout the central and peripheral nervous systems are referred to as the endocannabinoid system (ECS). In the central nervous system, ECs serve a number of important functions, such as the regulation of satiety, sleep, mood, and sensation of pain (Fride, 2005). In cutaneous nerves of humans and rodents, the ECS modulates both nociception and temperature sensation (Cravatt *et al.*, 2001; Potenzieri, Brink & Simone, 2009; Guindon *et al.*, 2011, 2013; Caterina, 2014). Only recently has a non-neuronal ECS been described in human skin, which also appears to regulate several key aspects of sebocyte biology (Maccarrone *et al.*, 2015; Oláh *et al.*, 2016; Zákány *et al.*, 2018).

SZ95 sebocytes produce the ECs AEA and 2-AG, and also express several receptors capable of binding cannabinoid compounds, including CB2 (Dobrosi *et al.*, 2008), TRPV1, TRPV2, and TRPV4 (Oláh *et al.*, 2014) (Table 4). Both AEA and 2-AG promote lipogenesis in SZ95 cells, and up-regulate PPARγ and cyclooxygenase-2 (COX2) expression through activation of mitogen-activated protein kinase (MAPK) signalling (Dobrosi *et al.*, 2008). Recently, it was shown that inhibition of endocannabinoid uptake by SZ95 sebocytes suppressed lipopolysaccharide-induced production of inflammatory cytokines (Zákány *et al.*, 2018). Overall, the function of the ECS in sebocytes appears to be to promote both lipogenesis and a shift towards a pro-inflammatory phenotype.

In contrast to ECs, several phytocannabinoids have been shown to negatively regulate lipid and interleukin production in SZ95 sebocytes (Oláh *et al.*, 2014, 2016). For example, cannabidiol (CBD) dose-dependently reverses AEA-induced lipogenesis in SZ95 cells and in organ-cultured human SGs. CBD also reduces lipid synthesis and production of inflammatory mediators elicited by incubation with arachidonic acid, linoleic acid, and testosterone (Oláh *et al.*, 2014). Importantly, these effects of CBD are mediated by a CB1/CB2-independent mechanism, through activation of TRPV4 vanilloid receptors (Oláh *et al.*, 2014).

Given that some of the observations summarised above were made in organ-cultured normal human skin, it is likely that the ECS also regulates human SG function *in vivo*. Yet, this remains to be formally demonstrated. Nevertheless, it has been reported that human SGs are immuno-reactive

Table 4. Expression of endocannabinoid system (ECS) components in sebaceous glands (SGs)

Component	Description	Species/ system	Methodology	Reference(s)
CB1	GPCR, cannabinoid receptor	Human SGs	IHC	Ständer *et al.* (2005); Dobrosi *et al.* (2008)
CB2	GPCR, cannabinoid receptor	Human SGs	IHC	Ständer *et al.* (2005); Dobrosi *et al.* (2008)
		SZ95 sebocytes	PCR	Dobrosi *et al.* (2008)
		Mouse SGs	IHC	Zheng *et al.* (2012)
TRPV1/2/4	Cation channels, vanilloid receptors	SZ95 sebocytes	PCR, WB	Oláh *et al.* (2014)
MGL	Monoacylglycerol lipase, breaks down 2-AG	Human SGs	IHC	Zákány *et al.* (2018)
		SZ95 sebocytes	PCR, WB	Zákány *et al.* (2018)
FAAH	Fatty acid amide hydrolase, breaks down AEA	Human SGs	IHC	Zákány *et al.* (2018)
		SZ95 sebocytes	PCR, WB	Zákány *et al.* (2018)
DAGLα/β	Diacylglycerol lipase, synthesises 2-AG	Human SGs	IHC	Zákány *et al.* (2018)
		SZ95 sebocytes	PCR, WB	Zákány *et al.* (2018)
NAPE-PLD	Implicated in AEA synthesis	Human SGs	IHC	Zákány *et al.* (2018)
		SZ95 sebocytes	PCR, WB	Zákány *et al.* (2018)

2-AG, 2-arachidonoylglycerol; AEA, anandamide; GPCR, G-protein-coupled receptor; IHC, immunohistochemistry; PCR, polymerase chain reaction; WB, western blot.

for both cannabinoid receptors, CB1 and CB2 (Ständer *et al.*, 2005; Dobrosi *et al.*, 2008). Furthermore, mouse and human SGs express several enzymes involved in endocannabinoid synthesis (Ma *et al.*, 2011; Zheng *et al.*, 2012; Zákány *et al.*, 2018) (Table 4).

Based on the SZ95 data described, it could be hypothesised that sebocyte-produced ECs work in an autocrine fashion to modulate both lipid synthesis and the production of inflammatory mediators. Whether such a system operates independently of the ECS that is present in cutaneous nerves is not known. The outcomes of current clinical trials (e.g. NCT03573518: *Evaluation of BTX 1503 in patients with moderate to severe acne vulgaris*) will soon provide clear indications as to the importance of the ECS in SG biology and pathology.

IX. MAJOR OPEN QUESTIONS AND TRANSLATIONAL RELEVANCE OF NEURO-REGULATION OF THE SEBACEOUS GLAND

The neurobiology and neuroendocrinology of SGs has, in the past, largely concerned only those with a professional interest in SG-associated diseases. However, research that has indicated several important functions of sebum (Schneider, 2015*a*; Dahlhoff *et al.*, 2016), will have widened the sphere of interest to clinicians and researchers concerned with broader aspects of skin disease and function. Moreover, the SG is an easily accessible and tractable entity, in which the fundamental aspects of mammalian tissue homeostasis can be studied (Hinde *et al.*, 2013).

It is possible, for example, to image murine SGs *in vivo*, in real time, and with the ability to resolve and track individual sebocytes (Pineda *et al.*, 2015). Furthermore, human SGs can be extracted from skin and maintained *ex vivo*, without dissociation into monolayer cultures (Ridden, Ferguson & Kealey, 1990; Guy, Ridden & Kealey, 1996; Langan *et al.*, 2015; Philpott, 2018). Such SG 'organ culture' can be used to dissect, without concern for potential systemic effects, the roles of any number of hormones, neuro-mediators, or small molecules, in directly regulating the turnover of this epithelial compartment.

Specifically, investigating the neuronal and neuroendocrine regulation of SGs promises to contribute further to fields of broad biological interest, such as maintenance of stem cell niches (Liao & Nguyen, 2014; Füllgrabe *et al.*, 2015; Kretzschmar *et al.*, 2016), tumour development and progression (Kyllo, Brady & Hurst, 2015; Quist *et al.*, 2016), lipid synthesis (Guy, Ridden & Kealey, 1996; Schneider & Paus, 2010; Jester, Potma & Brown, 2016), autophagy (Fischer *et al.*, 2017; Rossiter *et al.*, 2018), and interaction of the nervous system with cardinal signalling pathways in peripheral tissues, i.e. Hedgehog (Allen *et al.*, 2003; Brownell *et al.*, 2012), Wingless (Kretzschmar *et al.*, 2016; Lim *et al.*, 2016), and Hippo pathway signalling (Yosefzon *et al.*, 2018). In this review, we have presented the current state of knowledge of regulation of SG function by the nervous system. Where possible, mechanistic models and testable hypotheses have been presented, regarding how neurohormones and other nerve fibre-derived factors act to modulate the SG.

A common feature among several of the discussed pituitary neurohormones is that they are required for SGs to respond to testosterone (Ebling, Ebling & Skinner, 1969a; Petersen, Zone & Krueger, 1984; Akamatsu, Zouboulis & Orfanos, 1992; Fujie *et al.*, 1996; Rosignoli *et al.*, 2003) (Fig. 4), and thus for their continued maintenance (Ebling, 1957; Sweeney *et al.*, 1969; Thody & Shuster, 1971*b*; Chen, Belanger & Labrie, 1996; Azzi, El-Alfy & Labrie, 2006). So far, PRL (Ebling, Ebling & Skinner, 1969a; Ozegović & Milković, 1972), GH (Ebling, Ebling & Skinner, 1969a, 1969b; Ozegović & Milković, 1972; Ebling *et al.*, 1975a), and α-MSH (Thody & Shuster, 1975; Ebling *et al.*, 1975b; Thody *et al.*, 1976) have been demonstrated to exert this testosterone-sensitising ability on the cutaneous SGs of rats.

One hypothesis as to the sensitisation mechanism behind this phenomenon is, as mentioned, that these neurohormones regulate the enzymatic processing of testosterone to more active metabolites within the skin (Ebling *et al.*, 1971, 1973; Randall & Ebling, 1982; Ebling, 1986). Another possibility is that these neurohormones promote AR expression in sebocytes. Both scenarios may or may not involve the direct action of neurohormones on sebocytes themselves, although the reported expression of receptors for PRL (Choy, Nixon & Pearson, 1997; Craven *et al.*, 2001; Koizumi *et al.*, 2003), GH (Lobie *et al.*, 1990; Oakes *et al.*, 1992), and α-MSH (Chen *et al.*, 1997; Thiboutot *et al.*, 2000; Böhm *et al.*, 2002; Hong-Wei *et al.*, 2010), would suggest that the mode of action is, in fact, direct. Alternatively, these neurohormones might act on fibroblasts within the SG mesenchyme that signal to sebocytes *via* yet undescribed mechanisms. An alternative hypothesis, is that these neurohormones act on SG progenitor cells within the JZ. So far, none of these possibilities has been investigated.

Furthermore, several pituitary neurohormones can stimulate the SG independently of androgens. ACTH and α-MSH stimulate sebum production *in vivo*, in the absence of both testes and adrenal glands (Thody & Shuster, 1970*a*, 1975; Ozegović & Milković, 1972; Ebling *et al.*, 1975b; Thody et al., 1976; Thody, Meddis & Shuster, 1978) (Fig. 4). Again, the presence of cognate receptors for these neurohormones in the SG would strongly suggest that the action is exerted directly on sebocytes (Chen *et al.*, 1997; Thiboutot *et al.*, 2000; Böhm *et al.*, 2002; Hong-Wei *et al.*, 2010; Eisinger *et al.*, 2011; Zhang *et al.*, 2011) (Table 2; Fig. 6). Proof-of-principle, in this regard, can be seen when looking at the effects of the hypothalamic neurohormone CRH on sebocytes *in vitro*. CRH promotes lipid and cytokine production in SZ95 cells directly (Krause *et al.*, 2007; Ganceviciene *et al.*, 2009). Moreover, the synthesis of CRH (Krause *et al.*, 2007; Ganceviciene *et al.*, 2009), POMC (Kono *et al.*, 2001), and β-endorphin (Slominski, Paus & Mazurkiewicz, 1992; Furkert *et al.*, 1997; Mazurkiewicz, Corliss & Slominski, 2000) by sebocytes suggests the possible existence of autocrine pathways in sebocytes, utilising neurohormones that are typically associated with the neuroendocrine or nervous systems (Table 2; Fig. 6). Indeed, the HF is known to contain similar peripheral equivalents of central neuroendocrine axes, as mentioned previously (Paus *et al.*, 2014). Alternatively, some of these neurohormones might also be derived from cutaneous nerve endings (Roloff *et al.*, 1998; Slominski *et al.*, 1999). These peripheral 'neuro-autocrine' pathways may regulate several important aspects of follicular biology, including hair cycle progression, hair growth, pigmentation, and inflammation (Paus *et al.*, 2014).

However, given the evidently heavy dependence of the SGs on the presence of a functionally intact pituitary gland (Fig. 4), the function and importance of such peripheral neuroendocrine axes for SGs is as-yet unclear. One hypothesis proposes that external stressors may activate the peripheral neuroendocrine pathways in skin, which in turn activate the central neuroendocrine system, facilitating a systemic response to external stimuli (Jozic *et al.*, 2015). In any case, pharmacological manipulation of these pathways in cultured SGs or sebocyte cell lines should rapidly provide some indication of their importance relative to the central neuroendocrine system (Schneider & Zouboulis, 2018).

Despite the clear evidence demonstrating neurohormonal regulation of SGs, there is a relative dearth of data explicitly indicating that neurohormones contribute to the aetiology of acne. Increased acne prevalence among those suffering from neuroendocrine disorders like acromegaly is suggestive of a neurohormonal component to acne (Chalmers *et al.*, 1983; Jabbour, 2003), but it has so far been impossible to distinguish the role of the respective neurohormones from the fluctuating levels of androgens that are also observed in such conditions (Table 1). The link between seborrhoea and neuroendocrine dysfunction is more evident, especially in PD, where androgen levels are unchanged or even reduced (Ready *et al.*, 2004; Ruffoli *et al.*, 2008; Nitkowska *et al.*, 2015).

It is interesting to note, however, that isotretinoin reduces the systemic concentration of several pituitary neurohormones, which may in part explain its efficacy in treating acne (Karadag *et al.*, 2011, 2015). According to the inflammatory model of acne pathogenesis, CRH could represent an acnegenic neurohormone, having been demonstrated to promote neutral lipid synthesis and interleukin production *in vitro* (Krause *et al.*, 2007). Future research should aim to clarify the cellular targets of neurohormones within the PSU, and assess their ability to modulate sebaceous lipid synthesis and lipid composition, as well as the production of inflammatory mediators.

In addition to the neuroendocrine system, numerous clinical and experimental indicators suggest that the SG is also subject to control by the peripheral nervous system (Table 3). Despite this hypothesis having long been postulated, innervation of the SG has yet to be convincingly demonstrated. Indeed, direct innervation of the SG is only one of several possibilities (Fig. 10). The expression of nAChRα7 in mouse and human sebocytes (Proskocil *et al.*, 2004; Kurzen *et al.*, 2007; Li *et al.*, 2013b), and the lipogenic response of human sebocytes to ACh, would suggest that the SG can respond to nerve-derived factors directly (Li *et al.*, 2013b), although it may be possible that ACh *in vivo*, at least with regard to the SG, is derived from non-neuronal sources (Kurzen *et al.*, 2007; Wessler & Kirkpatrick, 2008) (Fig. 10).

A compelling alternative hypothesis is that nerve fibres instead contact and regulate SG progenitors, which are known to exist within the follicular compartment, rather than the SG itself (Nowak *et al.*, 2008; Jensen *et al.*, 2009; Barnes, Saurat & Kaya, 2017) (Fig. 10). The nerve supply to the isthmus region of the murine HF is well characterised, and resides in proximity to where SG progenitors are located (Botchkarev *et al.*, 1997; Bewick & Banks, 2016; Cheng *et al.*, 2018). Human HFs exhibit a very similar arrangement of follicular nerve fibres (Montagna & Parakkal, 1974*c*; Hordinsky & Ericson, 2008). The observed loss of expression of Lgr6 and Gli1 in the isthmus of mouse PSUs following denervation is particularly interesting (Brownell *et al.*, 2012; Liao & Nguyen, 2014), and may reflect a dependence of these cells on nerve-derived factors, such as Shh (Brownell *et al.*, 2012) (Fig. 10). However, a SG phenotype has yet to be described following cutaneous denervation. Should this model of neuronal regulation of SG progenitors prove to be accurate however, then the peripheral nervous system is potentially of great pathophysiological relevance; especially with regard to the 'comedo switch' hypothesis of acne pathogenesis, which postulates that comedones may form *via* abnormal differentiation of progenitor cells that supply the SG and infundibulum (Saurat, 2015; Clayton *et al.*, 2019).

Of course, neuronal regulation of the SG may occur by even more indirect means, perhaps through neuronal regulation of blood vessels in proximity to the SG, or even of cells within the CTS or dermis. A further complication can be found in the evidence suggesting that neuronal regulation of SG function may only occur as part of pathology. For example, sensitivity to botulinum toxin-A injection was only observed in those with abnormally high sebum production (Li *et al.*, 2013b), and peri-sebaceous SP immuno-reactive nerve fibres have only been reported in acne lesions (Toyoda *et al.*, 2002). SP, as well as endocannabinoids, may be neuromediators of special relevance to acne vulgaris, given the evidence for their pro-inflammatory effects on cultured sebocytes (Lee *et al.*, 2008). Another interesting, although completely unexplored, aspect of potential means of SG regulation is the interaction between the neuroendocrine and peripheral nervous systems. For example, PRL-R is known to regulate the sensitivity of nociceptive neurons in mice (Patil *et al.*, 2013). Perhaps a component of neurohormonal regulation of SGs is mediated through modulation of cutaneous nerve function.

When revisiting the matter of SG innervation, future researchers will have to keep in consideration the anatomical location of the PSUs, disease status, and hair cycle stage. Previous attempts to find direct innervation of the sebaceous epithelium may have been confounded by thin histological sectioning and a lack of broad-spectrum nerve fibre markers. With the advent of high-resolution confocal microscopy and means of staining and optically clearing entire tissues, it should now be possible to comprehensively and definitively map the nerves associated with rodent and human PSUs.

X. FINAL PERSPECTIVES

Evaluation of anti-acne therapies often uses reduced sebum production or SG involution as positive indicators of efficacy (Jalian *et al.*, 2015; Pan, Wang & Tu, 2017; Trivedi *et al.*, 2018), but these may not be desirable outcomes for maintenance of overall skin health, given the important physiological roles of sebum (Smith & Thiboutot, 2008; Dahlhoff *et al.*, 2016; Ehrmann & Schneider, 2016; Kobayashi *et al.*, 2019). Loss of sebum production might also be related to the pathogenesis of other dermatoses such as psoriasis and atopic dermatitis (van Smeden *et al.*, 2014; Shi *et al.*, 2015; Szántó *et al.*, 2019). Furthermore, it seems that certain components of sebum, as well as sebocyte-derived cytokines and inflammatory mediators, could exacerbate acne (Zouboulis, Jourdan & Picardo, 2014; Choi *et al.*, 2019) or perhaps contribute to comedo formation itself (Motoyoshi, 1983; Kim, Lee & Lee, 2018). Therefore, future research needs to identify signals and factors that can modulate sebum composition and sebocyte behaviour, rather than abrogate sebum production entirely (Törőcsik *et al.*, 2018; Zákány *et al.*, 2018). In this context, it may be possible, by manipulating the differentiation of progenitors for the infundibulum and SG (Jensen *et al.*, 2009; Page *et al.*, 2013; Barnes, Saurat & Kaya, 2017; Fontao *et al.*, 2017), to prevent comedo formation in acne without negatively affecting sebum secretion, or indeed, to specifically restore or modulate sebum production. To this end, the signalling pathways and stem cell dynamics underlying SG homeostasis must be thoroughly dissected.

The evidence reviewed above indicates that the nervous system is an important regulator of SG biology, able to modulate sebum production through either the direct action of neurohormones and nerve-derived factors, or by priming sebocytes to respond to other regulatory hormones, like androgens. However, much remains unknown about the specifics of these interactions of the neuroendocrine and peripheral nervous systems with SGs, especially with regard to how the indicated regulatory factors impact on sebum composition and other sebocyte-derived substances. Moreover, in most cases, the underlying molecular and cell biological mechanisms have yet to be identified.

XI. CONCLUSIONS

(1) Sebaceous glands (SGs) are mammalian epidermal appendages that produce lipids in the form of sebum. Sebum

serves important functions in cutaneous physiology, including waterproofing, maintaining skin barrier integrity, protection against ultraviolet radiation, anti-microbial defence, and thermoregulation.

(2) Acne vulgaris is associated with altered sebum composition; however, acne lesions might arise from abnormal differentiation of stem cells that supply the SG and infundibulum. Other inflammatory dermatoses, like psoriasis and atopic dermatitis, could potentially arise, or be exacerbated by, loss of the physiological functions of sebum.

(3) Neuroendocrine disorders often exhibit altered sebum levels. Addison's disease, acromegaly, growth hormone deficiency, and hyperprolactinaemia are all associated with abnormal sebum production.

(4) Pituitary neurohormones are required for normal SG function. Loss of the pituitary gland results in reduced sebum production, and reduced sensitivity of SGs to testosterone. Prolactin (PRL), growth hormone (GH), and α-melanocyte stimulating hormone (α-MSH) can partly restore normal sebum production when administered concomitantly with testosterone.

(5) Adrenocorticotropic hormone (ACTH) and α-MSH are able to stimulate sebum production independently of the adrenal glands and gonads, suggesting that they possess an ability to act on SGs directly. The hypothalamic neurohormone, corticotropin-releasing hormone (CRH), also exerts a direct effect on sebocytes, and may represent an autocrine neurohormone, produced by sebocytes to regulate lipid synthesis and cytokine production.

(6) Seborrhoea associated with Parkinson's disease (PD), phenylketonuria (PKU), anti-psychotics, and late pregnancy/lactation, may be a product of increased levels of systemic PRL. In PD and PKU, hyperprolactinaemia arises from loss of dopaminergic inhibition of PRL release, and pro-dopamine therapeutics reduce both PRL levels and sebum production. However, given that the skin itself can reportedly produce PRL, the extent to which local or systemic PRL is the key sebotrop(h)ic factor is unclear.

(7) GH may facilitate sebum production independently of testosterone, by up-regulating levels of insulin-like growth factor-1 (IGF-1). IGF-1 stimulates lipid synthesis and proliferation in sebocytes, and is also linked to acne vulgaris. The hypothalamic–pituitary–thyroid axis exerts a testosterone-independent sebotrophic effect through thyroid hormones.

(8) Peripheral nervous system disorders have been linked to aberrant sebum production in humans, and botulinum toxin has been used effectively to reduce sebum levels in seborrheic patients, indicating neuronal control of SG function. However, it remains unclear whether the SG is directly innervated by cutaneous nerves. Human sebocytes *in vitro* respond to several neurotransmitters, including acetylcholine and substance P (SP), which supports the hypothesis of direct innervation.

(9) An alternative model to that of direct SG innervation, is that cutaneous nerve fibres control the supply of cells to the SG, by innervating and regulating the sebaceous progenitor cells within the hair follicle (HF). The putative location of SG progenitors within the pilosebaceous unit (PSU) is densely innervated.

(10) Sebocytes synthesise and respond to endocannabinoids (ECs), which appear to act in an autocrine fashion to promote lipid synthesis and the production of inflammatory mediators. Phytocannabinoids, such as cannabidiol, ameliorate the pro-lipogenic and inflammatory effects of ECs *in vitro*. It is not yet known whether this sebaceous endocannabinoid system (ECS) operates independently of the peripheral nervous system.

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